



MSc Business Analytics Dissertation

Portfolio MGT7219

Detecting Fraudulent Job Postings Using Machine Learning
and Text Mining Techniques

Research Report

Name: Uzair Abdullah Khan

Student ID: 40385187

Word Count: 7827

Sept 2024

Queen's Business School

CANDIDATE DECLARATION

Declaration

This is to certify that:

- The dissertation comprises only my original work.
- Due acknowledgement has been made in the text to all other materials used.
- No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.



[Candidate's Signature]

Uzair Abdullah Khan

Printed Name

12/09/2024

Date

Acknowledgements

I would like to express my deepest appreciation to my supervisor, Dr. Rohit Nishant, whose expert guidance, thoughtful insights, and unwavering patience have been essential throughout the course of this research. His depth of knowledge and clarity of vision provided a solid framework that helped shape and refine this work, ensuring it met the highest academic standards. I feel privileged to have worked under his mentorship, which has been both a challenging and rewarding experience that has greatly contributed to my intellectual growth.

I also wish to extend my sincere thanks to Queen's Business School for fostering an environment of rigorous academic inquiry and promoting a culture of learning that encourages both creativity and critical thinking. The support provided by the faculty, particularly by our program director, Dr. Byron Graham, has been invaluable. His dedication to ensuring students have the necessary tools and support to succeed was instrumental in the completion of this project, and I am deeply grateful for his encouragement.

Lastly, I would like to acknowledge the immense support of my family and friends. Their encouragement, patience, and belief in my abilities were constant sources of strength throughout this journey. Their unwavering confidence in me provided the emotional and mental foundation necessary to see this research through to its conclusion, and for that, I am profoundly grateful.

Abstract

Fraudulent job postings pose significant risks to both job seekers and employers, contributing to financial loss, emotional distress, and reputational damage. In light of growing concerns regarding the prevalence of fake job postings, accurately identifying fraudulent job listings has become increasingly important. This research applies advanced machine learning techniques to detect fraudulent job postings, utilizing various models such as Decision Tree, Support Vector Machine, Random Forest, Naive Bayes, Gradient Boosting, and Logistic Regression. The analysis was conducted on a dataset of job postings, leveraging the TF-IDF method for text feature extraction and SMOTE for handling class imbalance.

The results reveal that the **Random Forest** classifier achieved the highest overall accuracy (97.06%) and a balanced performance in terms of precision, recall, and F1-score. However, the **Gradient Boosting** algorithm, while achieving comparable accuracy (96.92%), demonstrated better interpretability and robustness when handling complex patterns in the dataset. The inclusion of SMOTE significantly improved recall for the minority class but introduced a trade-off with precision, highlighting the challenges of handling imbalanced data in fraud detection. Models like **Naive Bayes**, while computationally efficient, showed lower recall for fraudulent postings, making them less effective in this context.

Several studies underscore the impact of fraudulent job postings. Fake job postings lead to financial exploitation and personal data theft for job seekers. Employers face reputational risks and decreased trust in recruitment platforms. These findings suggest that machine learning models could serve as a critical tool in mitigating these risks, offering a scalable solution to detect and prevent fraud in job postings. Policymakers and platform administrators can benefit from these findings by implementing stricter screening processes, enhancing platform security, and raising awareness about the consequences of fraudulent job postings.

Key Words:

- Fraudulent Job Postings
- Machine Learning
- SMOTE (Synthetic Minority Oversampling Technique)
- Text Classification
- Random Forest
- Gradient Boosting
- Support Vector Machine (SVM)

Table of Contents

Introduction	7
1.1 Overview	7
1.2 Business Problem.....	8
1.3 Purpose and Scope	9
Literature Review	9
2.1 The Importance of Identifying Fraudulent Job Advertisements	9
2.2 Key Features of Fraudulent Job Advertisements	10
2.3 Application of Machine Learning Techniques in Fraud Detection	10
Methodology.....	11
3.1 Data Understanding	11
3.2 Business Understanding	13
3.3 Data Preprocessing	14
3.3.1 Data Cleaning	14
3.3.2 Data Conversion	14
3.4 Model Selection and Training	15
3.4.1 Decision Tree Classifier	15
3.4.2 Support Vector Classifier (SVC).....	16
3.4.3 Random Forest Classifier	16
3.4.4 Naive Bayes Classifier	16
3.4.5 Gradient Boosting Classifier	17
3.4.6 Logistic Regression	17
3.5. Model Evaluation	18
3.5.1. Confusion Matrix.....	18
3.5.2. Classification Report.....	19
Results.....	19
4.1 Exploratory Data Analysis	19
4.2 Model Performance	22
4.2.1 Decision Tree Classifier	22
4.2.2 Support Vector Classifier (SVC).....	22
4.2.3 Random Forest Classifier	22
4.2.4 Naive Bayes Classifier	23
4.2.5 Gradient Boosting Classifier	23
4.2.6 Logistic Regression	23

4.3 Accuracy vs. Precision-Recall Trade-off:.....	24
4.4 Feature Importance Analysis of Gradient Boosting and Decision Tree Models.....	24
4.5 Cross Validation	25
4.6 Best Performing Result	25
4.7 Effectiveness and Impact of SMOTE	26
4.8 Business Oriented Findings.....	26
Discussion.....	26
5.1 Model Evaluation	27
5.1.1 Decision Tree Classifier.....	27
5.1.2 Support Vector Classifier (SVC).....	27
5.1.3 Random Forest Classifier	27
5.1.4 Naive Bayes Classifier	28
5.1.5 Gradient Boosting Classifier	28
5.1.6 Logistic Regression	28
5.2 Insights from Feature Importance in Fraud Detection	29
5.3 Improving Model Performance.....	29
Conclusion.....	30
Recommendations	31
References.....	32

Introduction

1.1 Overview

The rise of fraudulent job postings on the internet has posed serious risks to both job seekers and employers. Scammers increasingly exploit online job platforms to create deceptive advertisements that appear legitimate, making it difficult for individuals to distinguish between genuine and fraudulent job opportunities (Raza et al., 2022; Gurusami, 2018). These fraudulent schemes can result in severe financial, psychological, and reputational consequences for victims, emphasizing the need for robust detection mechanisms (Kassem, 2023).

This study aims to employ machine learning methods to enhance the accuracy of distinguishing between fraudulent and authentic job advertisements. Detecting fraudulent job postings involves analyzing large volumes of unstructured textual data, which can be achieved through advanced natural language processing (NLP) and machine learning (ML) techniques (Dastjerdi et al., 2019). By integrating interpretable machine learning models, this research seeks to provide valuable insights into the key features that influence the classification of job postings, enabling more transparent and explainable outcomes.

The dataset used for this study includes various text attributes related to job postings. Given the complexity of this data, the Cross Industry Standard Process for Data Mining (CRISP-DM) framework is adopted as the foundational methodology, guiding each phase of the research. A variety of ML algorithms, such as Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting Machines (GBM), Decision Trees (DT), and Logistic Regression, will be used to predict fraudulent job postings.

Previous studies have demonstrated that job posting fraud detection can be enhanced by using text-based features such as job descriptions, company profiles, and job requirements (Mehta et al., 2022). This research extends upon existing work by comparing several ML models and their effectiveness in detecting fraudulent job postings, while also addressing the limitations of black-box models through interpretable machine learning techniques. The results will provide valuable insights for improving the security and integrity of the online job market.

The primary contribution of this research is as follows:

1. Identify the machine learning model with the highest accuracy in classifying fraudulent job postings.
2. Determine the most important text features contributing to the detection of fraudulent job advertisements.
3. Examine the role of natural language processing in enhancing fraud detection accuracy.
4. Provide recommendations for improving job platform security and mitigating the risks posed by fraudulent job listings.

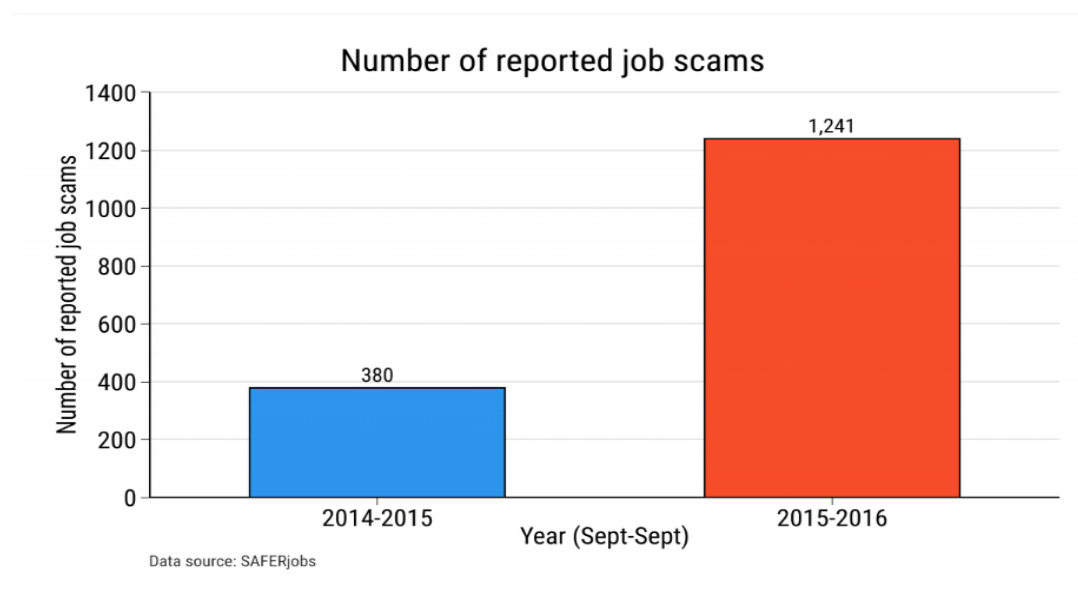
1.2 Business Problem

Fraudulent job postings pose a significant threat not only to job seekers but also to businesses and online job platforms. For job seekers, these deceptive ads can lead to financial loss, identity theft, and emotional distress. Globally, unemployment remains a pressing issue, with the International Labour Organization (ILO) estimating that over 207 million people were unemployed worldwide in 2022 (International Labour Organization, 2022). In the UK alone, as of 2023, unemployment stands at around 4.2% of the population, translating to approximately 1.42 million people actively seeking employment (Office for National Statistics, 2023). With thousands of job seekers relying on online job platforms for employment opportunities, the proliferation of fraudulent job listings presents a significant risk to both individuals and the broader job market.

The UK is home to numerous popular job platforms, including Indeed, Totaljobs, and Reed, which host millions of job listings daily. However, fraudulent job postings undermine the credibility of these platforms, leading to reputational damage for legitimate companies that advertise there (Baki et al., 2020). If users begin to associate a platform with scams, both the volume and quality of applicants can decrease, negatively impacting recruitment efforts for businesses. Moreover, platforms that fail to effectively screen fraudulent listings may face regulatory scrutiny, fines, or loss of business partners (Mutemi & Bação, 2023).

The economic implications are vast, as companies may need to allocate significant resources toward fraud prevention and security measures. A notable example occurred in 2021 when a UK-based job seeker fell victim to a fake job posting for a marketing role. The individual was defrauded out of £10,000 after providing sensitive banking information, which was then used for identity theft and financial fraud (BBC News, 2021). Such incidents highlight the potential harm fraudulent postings can cause to individuals and their trust in online platforms.

Failure to address these issues can lead to a loss of trust in job platforms, resulting in reduced user engagement and higher operational costs to regain credibility. Therefore, finding effective ways to detect and eliminate fraudulent job postings is critical for maintaining the integrity of job platforms and protecting both businesses and job seekers.



1.3 Purpose and Scope

The primary objective of this study is to evaluate various machine learning models and assess their ability to accurately detect fraudulent job postings. In doing so, this research seeks to uncover the most significant features within job advertisements that signal potential fraud. Five machine learning algorithms—Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting (GB), Decision Trees (DT), and Logistic Regression—will be evaluated for their classification performance. Additionally, natural language processing (NLP) techniques will be integrated to enhance model accuracy by extracting valuable insights from job posting text data.

Given the importance of interpretability in fraud detection models, Decision Trees and Gradient Boost will be employed to analyze feature importance and clarify the contributions of specific text features in distinguishing between fraudulent and authentic job postings.

The dataset contains thousands of records with features such as job title, company profile, description, requirements, and benefits. By applying data cleaning and pre-processing techniques, the dataset will be prepared for model training and evaluation. Various statistical evaluation metrics, such as accuracy, precision, recall, and F1-score, will be used to compare the performance of the machine learning models.

The findings from this research will inform policymakers and platform administrators about the most effective strategies for detecting fraudulent job postings and improving the integrity of the online job market. Moreover, this study aims to contribute to a safer and more secure employment ecosystem by providing data-driven recommendations based on the predictive performance of advanced machine learning models.

The rest of this paper is structured as follows: The 'Literature Review' chapter discusses existing research on job posting fraud detection and relevant machine learning techniques. The 'Methodology' chapter outlines the proposed machine learning models, data collection, pre-processing, and evaluation methods. The 'Findings' chapter presents the model performance results and feature importance analysis. The 'Discussion' chapter provides an interpretation of the results within the broader context of job fraud detection literature. Lastly, the 'Conclusion' chapter summarizes the key findings, and the 'Recommendations' chapter addresses the study's limitations and future research opportunities.

Literature Review

This chapter reviews the existing research on detecting fraudulent job advertisements using various machine learning (ML) techniques and methodologies. The literature is organized into three key sections. First, it discusses the importance of detecting fraudulent job postings and their impact on job seekers and employers. Second, it examines the linguistic and structural features that distinguish fake job postings from legitimate ones. Lastly, it explores the application of ML techniques in fraud detection and evaluates their effectiveness.

2.1 The Importance of Identifying Fraudulent Job Advertisements

The increasing prevalence of fraudulent job advertisements presents significant risks to both job seekers and employers. These fraudulent ads exploit the desperation of vulnerable job seekers, often

those seeking remote work or experiencing financial instability. Victims may experience financial loss, identity theft, and emotional distress due to these scams. Detecting fraudulent postings is crucial for protecting job seekers and maintaining the credibility of online job platforms. Research highlights that job seekers are particularly vulnerable to scams that promise unrealistic salaries or ask for sensitive personal information at the early stages of the recruitment process (Ravenelle et al., 2022; Praveen et al., 2023; Rofik, 2024). Failing to detect fraudulent postings can lead to reputational damage for job platforms and decrease user trust in the legitimacy of job opportunities, which can in turn weaken the job market (Srivastava, 2022; Jiang, 2022; Hanif, 2024).

Employers also face challenges from fraudulent job advertisements. When legitimate companies advertise on platforms that fail to screen fraudulent listings effectively, it can harm their brand image, making it difficult to attract qualified candidates. Users may become wary of the legitimacy of job ads, ultimately harming recruitment efforts. The rise in fraudulent postings is a growing concern for both job platforms and legitimate businesses that depend on these platforms to connect with potential employees. Therefore, having robust fraud detection mechanisms in place is critical to preserving the integrity of online job platforms (Sandhu, 2019; Sipayung et al., 2023).

2.2 Key Features of Fraudulent Job Advertisements

Several studies have identified distinct linguistic and structural markers that are characteristic of fraudulent job advertisements. Fraudulent ads often exhibit vague or ambiguous job descriptions, unrealistic salary promises, and minimal details about the job role or employer (Lukersmith et al., 2021). For example, job ads that promise quick and easy income with little work required are frequently associated with scams. Additionally, these ads may request personal information such as Social Security numbers or banking details early in the recruitment process, a major red flag for job seekers (Ullah & Jamjoom, 2023; Grant-Smith, 2022).

The language used in fraudulent job advertisements can also differ significantly from that in legitimate postings. Fraudulent ads are more likely to include grammatical errors, misspellings, and informal tone, which can undermine their professionalism (Kudo et al., 2019). Vague terminology and frequent use of generic words such as "work," "earn," or "opportunity" are other indicators of fraud (Feng et al., 2018). Text mining and natural language processing (NLP) techniques have proven effective in identifying these linguistic patterns, enabling automated systems to flag potentially fraudulent postings for further review.

2.3 Application of Machine Learning Techniques in Fraud Detection

Machine learning has emerged as a powerful tool for detecting fraudulent job advertisements. Various algorithms, including logistic regression, decision trees, support vector machines (SVM), and random forest classifiers, have demonstrated strong performance in classifying job ads as either legitimate or fraudulent based on text-based features according to empirical analysis conducted by Lin (2023). These ML models are often combined with natural language processing (NLP) techniques to analyse the structure, tone, and content of job advertisements.

Logistic regression and SVM have been particularly effective in binary classification tasks, such as distinguishing between fraudulent and legitimate job ads (Shim & Seok, 2012). Logistic regression models the probability of fraud based on various input features, while SVM constructs a hyperplane

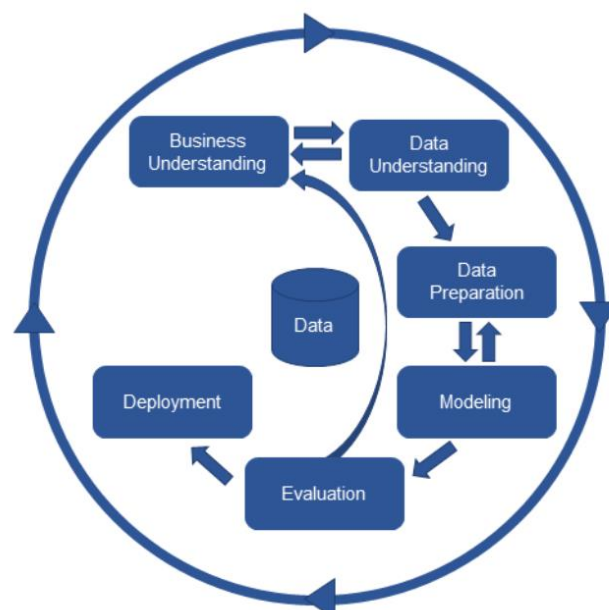
to separate the two classes. Both methods have been shown to achieve high levels of accuracy in fraud detection (Bekkerman & Gavish, 2011).

Decision trees and random forests have also proven to be valuable in detecting fraudulent job postings, particularly when combined with text mining techniques. Decision trees build a tree-like structure by making decisions based on input features, such as job description length, salary range, and job requirements. Random forests, an ensemble of decision trees, further enhance accuracy by reducing overfitting and improving the model's robustness (Assiri et al., 2020; , Mijwil & Abttan, 2021). Ensemble methods like random forests and Gradient Boosting Machines (GBM), which combine multiple weak classifiers to produce a more accurate model, have demonstrated significant improvements in fraud detection accuracy.

Recent advancements in machine learning have introduced more sophisticated algorithms, such as deep learning and neural networks, which show great promise in detecting fraudulent job ads. These techniques, especially when combined with NLP, can capture complex patterns in text data and deliver high accuracy (Kim et al., 2019). However, these models are often considered "black boxes," meaning their decision-making process is difficult to interpret (MoDastoni, 2023). While neural networks have the potential to improve fraud detection, the lack of interpretability can pose challenges for businesses and policymakers who need transparency in decision-making systems.

Methodology

This section provides a detailed explanation of the entire coding process, aligned with the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. It covers the understanding and preprocessing of data, modelling steps, and evaluation. The CRISP-DM methodology ensures a structured approach, guiding each phase of the project from business understanding to data preparation and model evaluation. The goal is to explain the methodology behind the classification of fraudulent job postings using various machine learning models. The code is structured in several steps, covering data preprocessing, model selection, evaluation, and the application of SMOTE to handle class imbalance.



3.1 Data Understanding

The dataset for this study was sourced from Kaggle and was uploaded in 2020. It contains approximately 18,000 job postings, of which about 800 are labelled as fraudulent. The dataset originated from the Aegean website, which collected real-world job postings and includes various text-based and meta-data features related to those job listings. The dataset is highly imbalanced, with real job postings significantly outnumbering fraudulent ones at an approximate ratio of 20:1, which presents a challenge in accurately identifying the minority class (fraudulent postings).

The dataset contains several important features, both text-based and categorical, that provide detailed information about each job listing. Key columns include *job_id*, which uniquely identifies each job posting, and *title*, which offers insight into the role being advertised. Location information is captured in the *location* column, although some values may be incomplete or missing. *Department* indicates the company department, but it has many missing values, and *salary_range* provides expected salary details, though it is inconsistently filled. *Company_profile* describes the company, while *description* outlines job responsibilities and *requirements* summarizes necessary skills and qualifications. The *benefits* column details perks offered by the company. *Telecommuting* is a binary variable showing whether remote work is allowed and *has_company_logo* indicates if a company logo is present in the job post. Similarly, *has_questions* is a binary indicator of qualifying questions for applicants. *Employment_type* categorizes the type of employment (e.g., full-time or part-time), while *required_experience* and *required_education* provide details about experience and education levels needed. *Industry* refers to the sector to which the job belongs, and *function* indicates the role (e.g., marketing). The target variable, *fraudulent*, signifies whether a job posting is fraudulent (1 = yes, 0 = no). Together, this structured data (e.g., *telecommuting*, *has_company_logo*) and unstructured text data (e.g., *company_profile*, *description*, *requirements*) offer a robust dataset for fraud detection.

Column Name	Character Type	Description
job_id	int	Unique identifier for each job posting.
title	text	The title of the job posting, providing insight into the advertised role.
location	text	The city, state, or country where the job is located (some values may be incomplete).
department	text	The department within the company (many missing values).
salary_range	text	The expected salary range for the role (often missing or inconsistently filled).
company_profile	text	A description of the company posting the job.
description	text	Detailed job responsibilities and requirements.
requirements	text	A summary of skills, experience, and qualifications required for the job.
benefits	text	Information about the perks and benefits offered by the company.
telecommuting	int	A binary variable indicating if the job allows telecommuting (1 = yes, 0 = no).
has_company_logo	int	A binary indicator of whether the job posting includes a company logo (1 = yes, 0 = no).
has_questions	int	A binary indicator of whether the job posting includes qualifying questions (1 = yes, 0 = no).
employment_type	text	The type of employment (e.g., full-time, part-time, contract).
required_experience	text	The level of experience required for the role (e.g., mid-level, associate).
required_education	text	The minimum education requirement for the job (e.g., high school, bachelor's degree).
industry	text	The sector or industry to which the job belongs (e.g., IT, finance).
function	text	The job function (e.g., marketing, sales).
fraudulent	int	Target variable indicating whether the job posting is fraudulent (1 = yes, 0 = no).

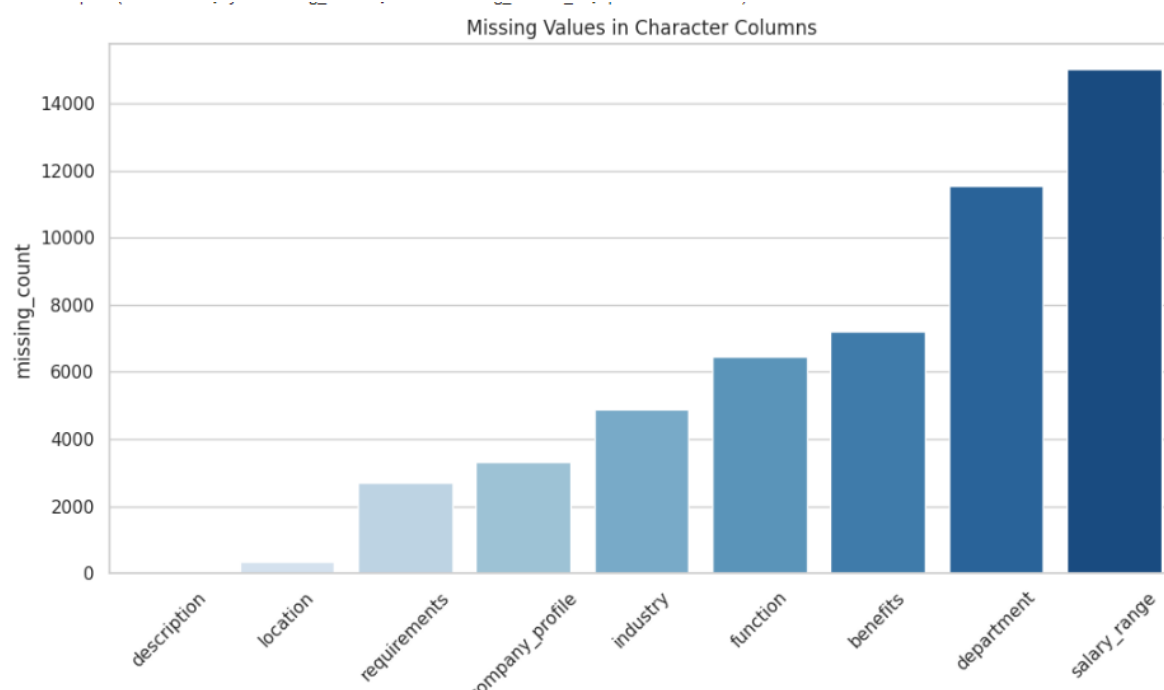
3.2 Business Understanding

The choice of machine learning models in this study is driven by essential business goals, with particular attention to precision and recall. From a business perspective, high recall is critical because it ensures the detection of as many fraudulent job postings as possible, protecting users from scams and safeguarding the platform's reputation (Kharraz et al., 2018). However, focusing solely on recall can lead to a higher number of false positives, where legitimate job postings are mistakenly flagged as fraudulent (McCann et al., 2002). This is a significant concern for businesses, as wrongly rejected listings can harm legitimate employers and diminish the platform's credibility. Therefore, precision becomes equally important, ensuring that legitimate job listings are not unduly rejected. Balancing recall and precision is crucial for maintaining user trust and business sustainability, as businesses must minimize false positives without compromising on the need to protect against fraud (Kadam, 2024). By carefully selecting and evaluating models, the goal is to develop an effective fraud detection system that aligns with the business objective of maximizing security while minimizing operational disruptions.

3.3 Data Preprocessing

3.3.1 Data Cleaning

Several features in the dataset contain a significant number of missing values that were addressed during the data cleaning process. The columns *salary_range*, *department*, and *benefits* have more than 70% missing data and were excluded from the analysis to avoid bias and ensure the integrity of the dataset. Additionally, *location* and *company_profile* contain missing values but were retained because they can potentially provide valuable insights into distinguishing between real and fake job postings. The *required_experience* and *required_education* columns also contain missing data but were kept as they offer important details about job listings and can improve model performance. The missing values in these columns were handled using appropriate imputation techniques, such as replacing them with the mode or using placeholder text for missing descriptions.



For textual features like *description* and *company_profile*, basic text cleaning techniques were applied, including removing HTML tags, punctuation, and unnecessary spaces. Tokenization, stemming, and stopwords removal were also applied to ensure consistency and reduce noise in the text data.

3.3.2 Data Conversion

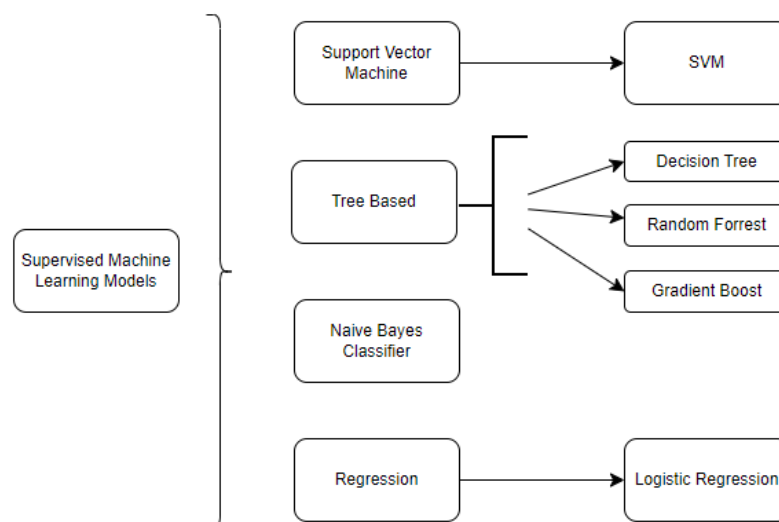
For categorical variables such as *telecommuting*, *has_company_logo*, *has_questions*, *employment_type*, *industry*, and *function*, conversion to numerical formats was necessary for machine learning models. Variables like *telecommuting*, *has_company_logo*, and *has_questions* are already binary and were used directly in the models.

The categorical variables *employment_type*, *required_experience*, *required_education*, *industry*, and *function* were converted to numerical form using TF-IDF to capture their non-ordinal nature without introducing unnecessary assumptions about the relationships between categories.

In the case of text-based features, such as *description*, *company_profile*, and *requirements*, the textual data was vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) to convert text into numerical vectors. This approach captures the importance of words within the corpus and helps models identify relevant patterns in the job postings. By converting these textual features into numerical formats, the models are able to understand and process the complex data more effectively.

These data cleaning and conversion steps are crucial for preparing the dataset for the modelling process, ensuring that missing values are appropriately handled, and categorical and textual data are converted into a format suitable for machine learning algorithms (Erol et al., 2022).

3.4 Model Selection and Training



3.4.1 Decision Tree Classifier

The Decision Tree classifier splits the data into subsets based on feature values, forming a tree-like structure where each node represents a decision. Each branch represents a choice, and the leaf nodes signify the classification outcomes. The model recursively partitions the data to minimize the variance in the target variable within each subset, creating a robust yet interpretable classification system.

Why Decision Tree?

- **Simplicity and Interpretability:** Decision Trees are easy to understand and visualize, making them one of the most interpretable models. Their transparent structure allows clear insight into how decisions are made based on feature importance (Blake et al., 2012).
- **Handling Mixed Data Types:** They can manage both numerical and categorical data efficiently, making them versatile for different datasets.
- **Feature Importance:** Decision Trees can highlight the most important features in classification tasks, offering insights into which features are the strongest predictors (Jiang, 2024).

With and Without SMOTE: This model is evaluated both with and without SMOTE to assess how oversampling affects performance on imbalanced data, particularly in terms of recall for the minority class (fraudulent jobs).

3.4.2 Support Vector Classifier (SVC)

Support Vector Classifiers (SVC) find a hyperplane that best separates different classes in a high-dimensional feature space. This hyperplane is optimized to maximize the margin between classes, ensuring that the classifier can generalize well to unseen data.

Why SVC?

- **High Dimensionality:** SVCs perform well with high-dimensional datasets, such as text data represented using TF-IDF vectors. Their ability to create decision boundaries in higher dimensions makes them suitable for tasks like text classification.
- **Binary Classification:** The linear kernel in SVC is especially effective for binary classification tasks, making it a good fit for distinguishing between legitimate and fraudulent job postings (Sitepu, 2022).
- **Robustness to Overfitting:** SVCs can generalize well, even when the data has many features compared to the number of samples (Han & Jiang, 2014).

With and Without SMOTE: Like the Decision Tree, SVC is evaluated with SMOTE to determine its performance in detecting fraudulent jobs in an imbalanced dataset.

3.4.3 Random Forest Classifier

Random Forest is an ensemble learning technique that constructs multiple decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees.

Why Random Forest?

- **Ensemble Learning:** By aggregating the results of multiple decision trees, Random Forests improve prediction accuracy and reduce overfitting, making them highly robust (Biau & Scornet, 2016).
- **Handling High-Dimensional Data:** Random Forests can efficiently handle datasets with a large number of features, such as text-based datasets where features are created through TF-IDF (Turner et al., 2017).
- **Feature Importance:** Random Forest provides valuable feature importance scores, which are useful for identifying the most significant predictors in detecting fraudulent jobs.

With and Without SMOTE: Random Forests are evaluated with SMOTE to see if oversampling the minority class improves recall while balancing precision.

3.4.4 Naive Bayes Classifier

Naive Bayes classifiers are probabilistic models that apply Bayes' theorem with the assumption of independence between predictors. In text classification, they are particularly effective due to their simplicity and the ability to handle large feature spaces (Wibawa et al., 2019; Chen & Fu, 2018).

Why Naive Bayes?

- **Efficient for Text Classification:** Naive Bayes works well with text data, where the assumption of independence between words (features) is often justified (Fernando et al., 2019).
- **Speed and Scalability:** It is computationally efficient, making it suitable for large datasets with high-dimensional feature spaces.
- **Well-Suited for NLP:** Due to its efficiency in handling large vocabularies, Naive Bayes is commonly used for tasks such as spam detection, making it a natural fit for detecting fraudulent job postings (Zhang, 2016).

Without SMOTE: Naive Bayes was not evaluated with SMOTE because the model relies on probability distributions, which may not function well with synthetic data.

3.4.5 Gradient Boosting Classifier

Gradient Boosting builds an ensemble of weak learners (typically decision trees) sequentially, where each new tree tries to correct the errors made by the previous trees. This approach allows Gradient Boosting to iteratively improve its predictions, especially on challenging datasets (Bakasa, 2023).

Why Gradient Boosting?

- **High Predictive Power:** Gradient Boosting excels in capturing complex patterns in the data by focusing on reducing the errors made in previous iterations, making it highly effective for imbalanced datasets like fraudulent job detection (Smiti et al., 2022).
- **Flexibility:** It can handle both classification and regression tasks, and it is known to perform well in many machine learning challenges and competitions (Xia et al., 2023).
- **Handling Imbalance:** Gradient Boosting is particularly well-suited for imbalanced datasets, as it places more focus on the difficult-to-predict instances, such as fraudulent job postings.

With and Without SMOTE: SMOTE is applied to Gradient Boosting to further improve its recall for fraudulent postings, helping to ensure that minority class samples are effectively identified.

3.4.6 Logistic Regression

Logistic Regression is a fundamental statistical method used for binary classification. It models the probability of a binary outcome based on one or more predictor variables using the logistic function (Rathi et al., 2010).

Why Logistic Regression?

- **Simplicity and Interpretability:** Logistic Regression is easy to implement and interpret, making it an ideal baseline model. Its simplicity allows for straightforward understanding of how each feature impacts the classification.
- **Effective for Large Datasets:** Logistic Regression performs well with large datasets and can handle text data effectively, especially when paired with regularization techniques to prevent overfitting (Kucuk, 2024).

- **Baseline Model:** Logistic Regression provides a solid baseline for comparison with more complex models. It is often used as the first model in binary classification tasks due to its simplicity and effectiveness (Rahman et al., 2021).

Without SMOTE: Logistic Regression was not used with SMOTE in this case because it already serves as a robust baseline, and SMOTE is typically more beneficial for more complex models where imbalance significantly affects performance.

3.5. Model Evaluation

Metrics	Definition	Formula	Evaluation/Range
Accuracy	Accuracy measures the proportion of correctly classified instances (both real and fraudulent jobs) out of all instances.	$\frac{TP+TN}{TP+TN+FP+FN}$	Values range from 0 to 1. A value closer to 1 indicates higher accuracy. However, it can be misleading in imbalanced datasets.
Precision (Class 1)	Precision is the proportion of true positive predictions (fraudulent jobs) among all positive predictions (both true and false).	$\frac{TP}{TP+FP}$	Precision ranges from 0 to 1. A high value means that most predicted fraudulent jobs are indeed fraudulent. It's important in reducing false positives.
Recall (Class 1)	Recall (also called Sensitivity) measures the proportion of actual positives (fraudulent jobs) that were correctly predicted.	$\frac{TP}{TP+FN}$	Recall ranges from 0 to 1. High recall ensures that most fraudulent jobs are detected. This is crucial when missing fraudulent jobs is a serious issue.
F1-Score (Class 1)	The F1-score is the harmonic mean of Precision and Recall. It balances the two metrics, useful when there is a tradeoff.	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	F1-score ranges from 0 to 1. A value closer to 1 indicates a good balance between Precision and Recall, suitable for imbalanced datasets like fraud detection.
Confusion Matrix	The Confusion Matrix provides a complete view of prediction performance by showing counts of TP, TN, FP, and FN.	$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$	It helps evaluate performance beyond simple accuracy, highlighting where the model is making errors (false positives or false negatives).

3.5.1. Confusion Matrix

The confusion matrix provides insight into the true positives, true negatives, false positives, and false negatives. This helps us understand how well the model distinguishes between fraudulent and non-fraudulent job postings (Faraji, 2022; Hansraj et al., 2021).

3.5.2. Classification Report

The classification report provides key metrics such as precision, recall, and F1-score. These metrics help evaluate the model's performance in handling the minority class (fraudulent jobs):

- **Precision:** The percentage of correctly predicted fraudulent jobs out of all predicted fraudulent jobs.
- **Recall:** The percentage of correctly predicted fraudulent jobs out of all actual fraudulent jobs.
- **F1-score:** The harmonic mean of precision and recall, giving a better overall measure of a model's performance with imbalanced data.

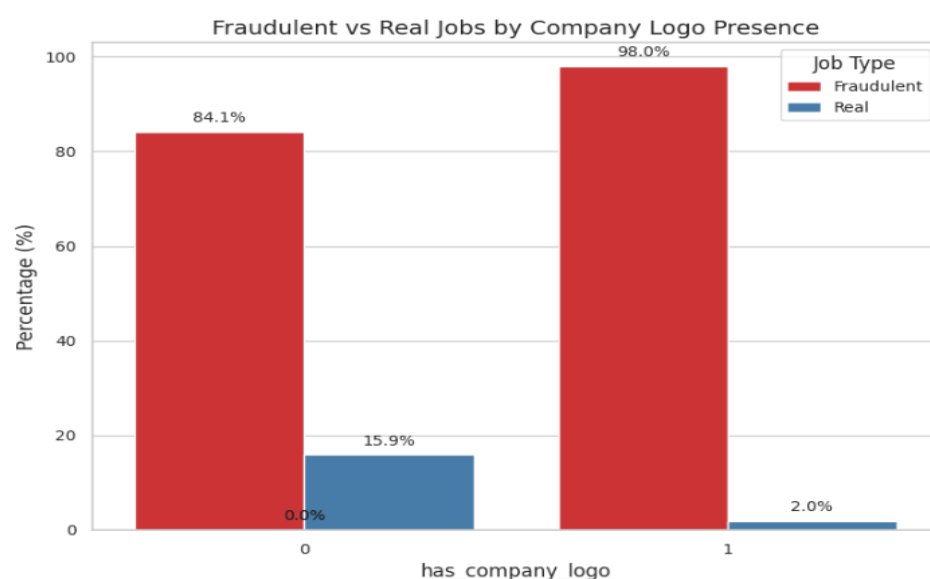
Results

This chapter outlines the results of the machine learning models used to identify fraudulent job postings. First, the data was cleaned and preprocessed to address missing values and prepare features for machine learning models. Second, the performance of the models was evaluated using accuracy, precision, recall, F1-score, and confusion matrix. Third, SMOTE was applied to handle class imbalance, and the impact on model performance was analyzed. Finally, suggestions for improving model performance were considered.

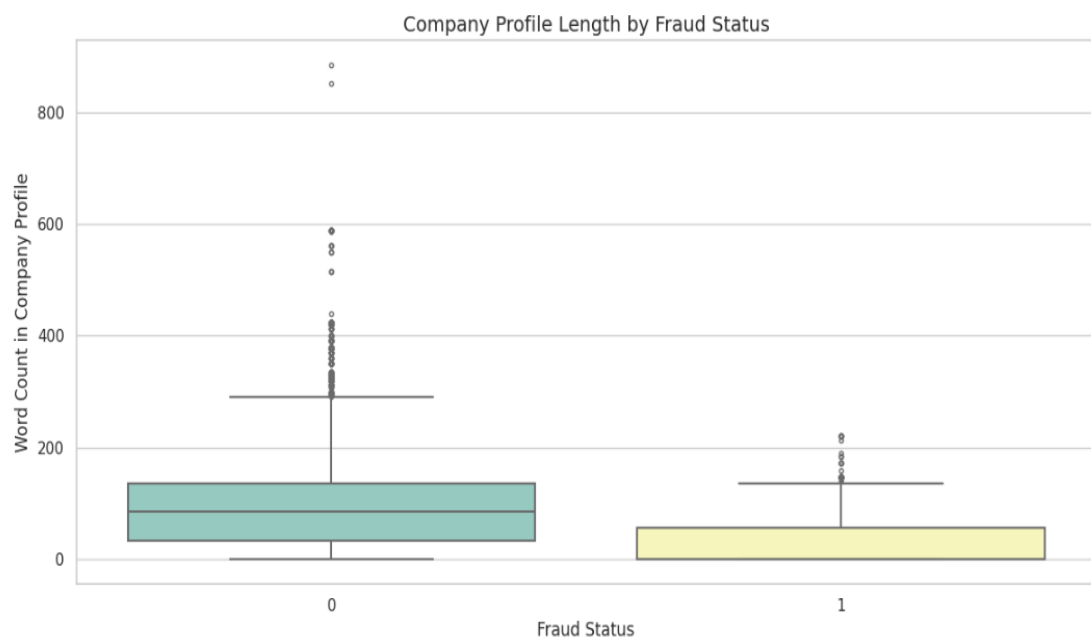
4.1 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to understand the distribution of the features and their relationship with the target variable, fraudulent. The key features analysed include *has_company_logo*, *company_profile*, *description*, and *required_education*. Insights were gained into how these features vary between fraudulent and non-fraudulent job postings:

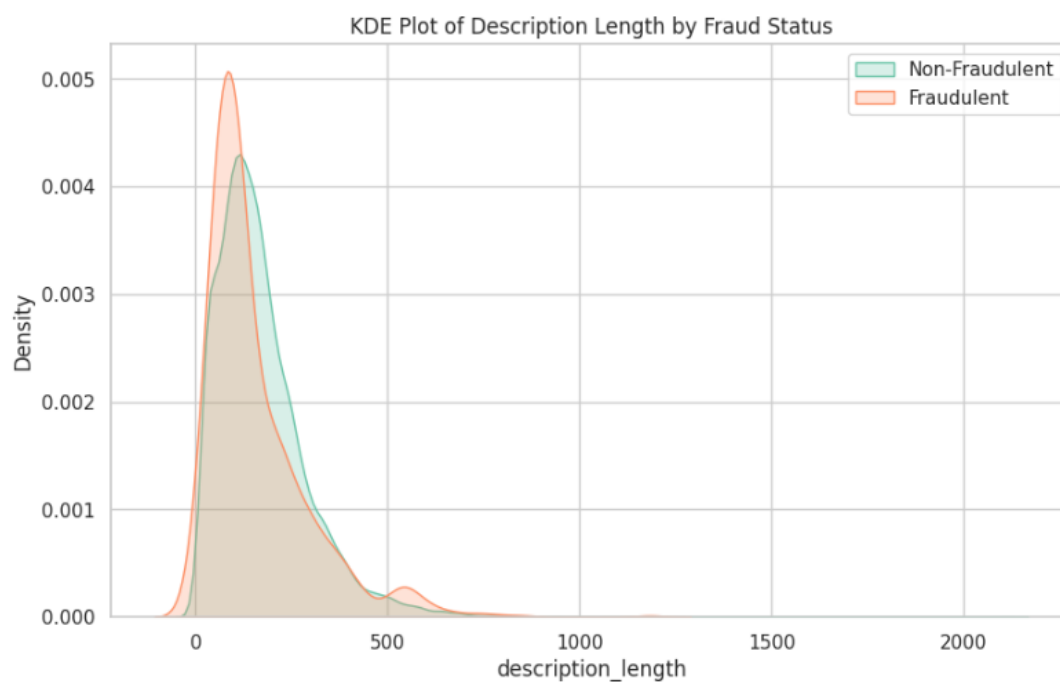
- **Has Company Logo:** The probability of jobs to be fraudulent is eight times more likely in case the company logo is not present than when it is present. In fact, more than three quarters of the jobs posted in the data set did not have any company logo.



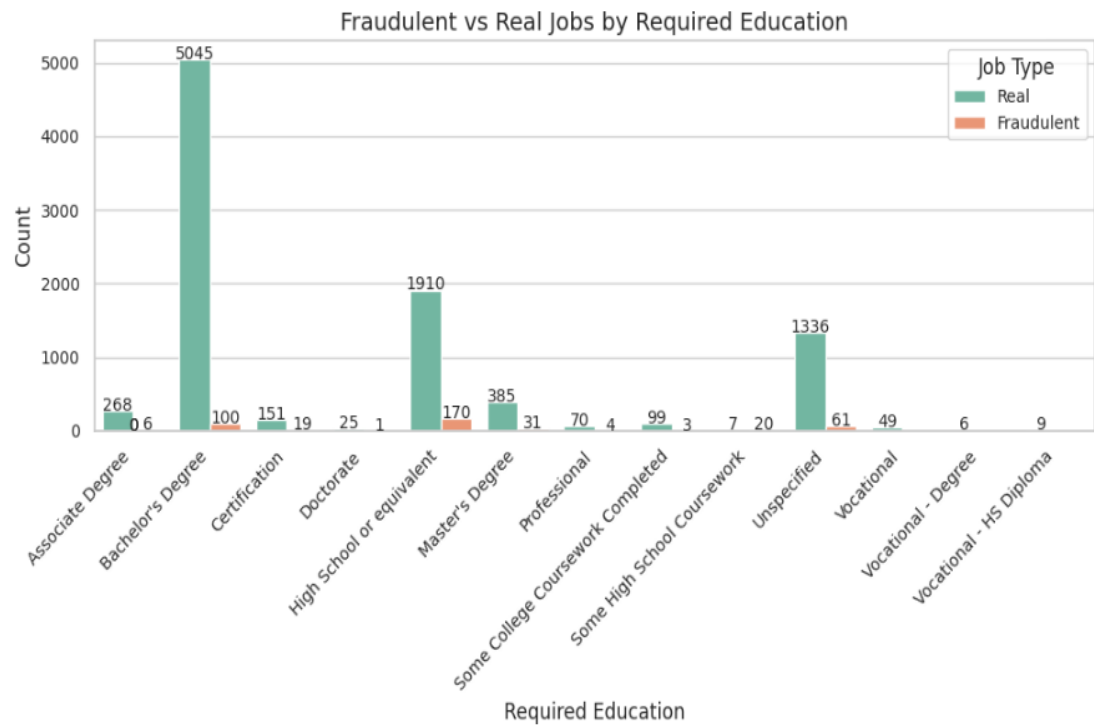
- **Company Profile:** Authentic job postings tend to have detailed company profiles, while fraudulent job postings often have generic or missing descriptions. Also, the average word count of company profiles is considerably larger in real jobs compared to fake ones.



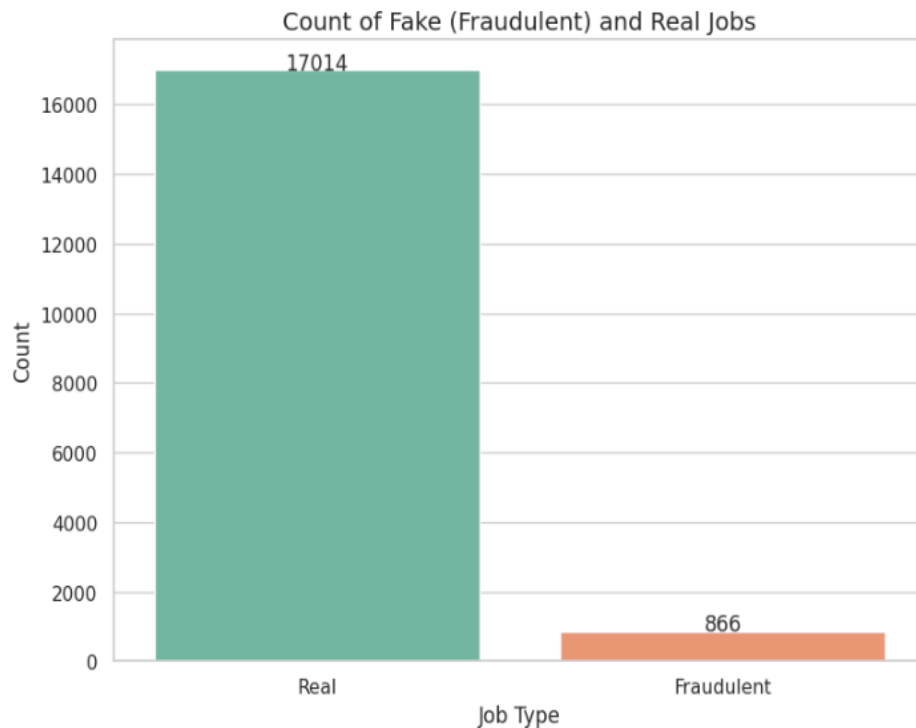
- **Job Description:** Fraudulent jobs typically have vague job descriptions or offer unrealistic benefits and perks, which helps identify fraud. Similar to the company profile, the jobs which are fake tend to have smaller word count compared to those with the real jobs.



- **Required Education:** Most fraudulent postings either leave this field blank or specify minimal educational requirements (Bachelor's degree/High School) compared to genuine postings, which tend to be more specific.



- **Data Set Imbalance:** The dataset was highly imbalanced, with only around 800 fake job postings out of a total of nearly 18,000 postings, reinforcing the need for oversampling methods like SMOTE to handle class imbalance.



4.2.4 Naive Bayes Classifier

The Naive Bayes classifier had the lowest recall (10%) for fraudulent job postings without SMOTE, with an accuracy of 95.44%. The model was unable to capture most fraudulent jobs, although it had reasonable precision (0.71). SMOTE was not applied to Naive Bayes due to its reliance on probabilities, which does not perform well with synthetic data.

- **Without SMOTE:** Struggled to detect fraudulent jobs, making it unsuitable for this task.

4.2.5 Gradient Boosting Classifier

Gradient Boosting provided one of the best performances without SMOTE, with an accuracy of 96.92% and a balanced F1-score (0.54) for fraudulent job detection. After SMOTE, the recall increased to 99%, but precision dropped to 0.22.

- **Without SMOTE:** Good balance between precision and recall.
- **With SMOTE:** High recall but a significant drop in precision.

4.2.6 Logistic Regression

Logistic Regression achieved an accuracy of 96.39% without SMOTE and high precision (1.00), but it had a low recall of 23% for fraudulent jobs. This model performed similarly to the other models with high accuracy but struggled to detect fraudulent postings.

- **Without SMOTE:** High precision but missed many fraudulent jobs.
- **No SMOTE Applied.**

Model	SMOTE	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-score (Class 1)
Decision Tree	No	97.01%	0.97	0.39	0.56
Decision Tree	Yes	84.48%	0.24	0.99	0.38
Support Vector Classifier	No	96.87%	1.00	0.35	0.52
Support Vector Classifier	Yes	84.48%	0.24	0.99	0.38
Random Forest	No	97.06%	1.00	0.39	0.56
Random Forest	Yes	84.65%	0.24	0.99	0.38
Naive Bayes	No	95.44%	0.71	0.10	0.17
Naive Bayes	-	-	-	-	-
Gradient Boosting	No	96.92%	0.98	0.37	0.54
Gradient Boosting	Yes	82.63%	0.22	0.99	0.36
Logistic Regression	No	96.39%	1.00	0.23	0.38

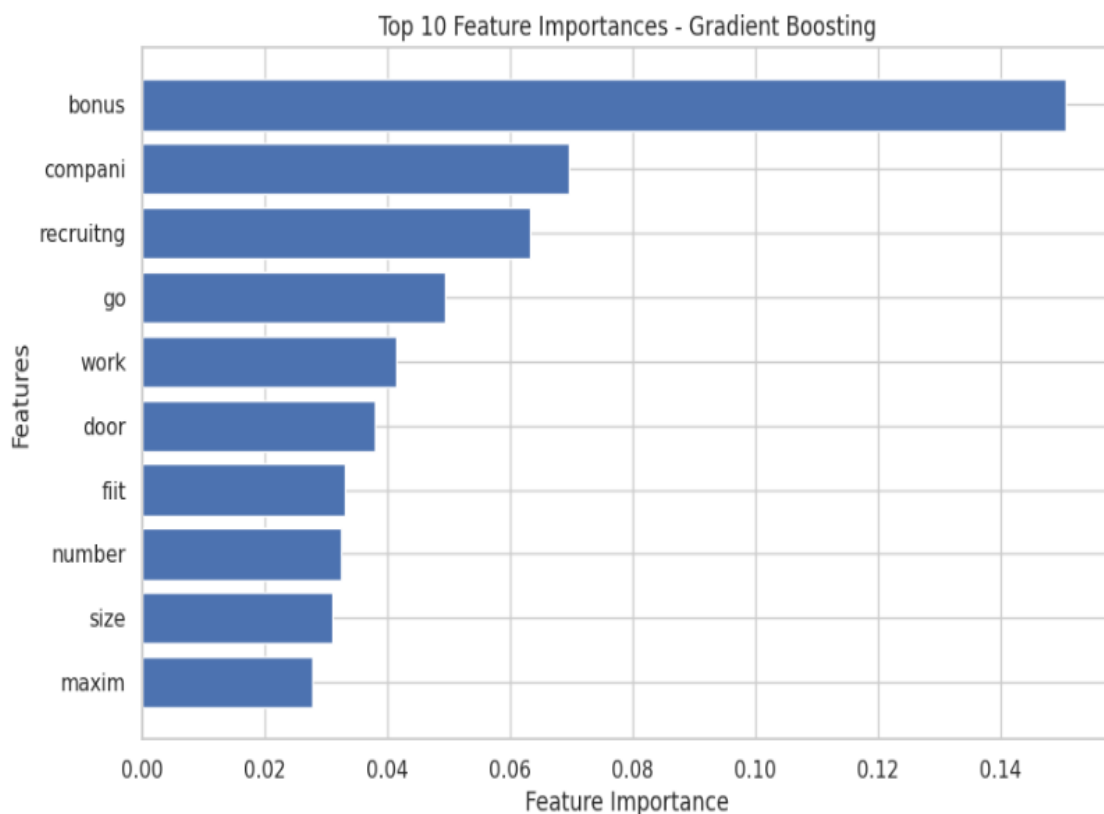
4.3 Accuracy vs. Precision-Recall Trade-off:

The results show that most models without SMOTE achieve high overall accuracy, exceeding 95%. However, these models struggle with recall for the minority class—fraudulent job postings—meaning that they miss a significant number of these fraudulent ads (Kh.TQ, 2024). For instance, models like the Decision Tree and SVM performed well overall but had low recall rates for fraudulent jobs.

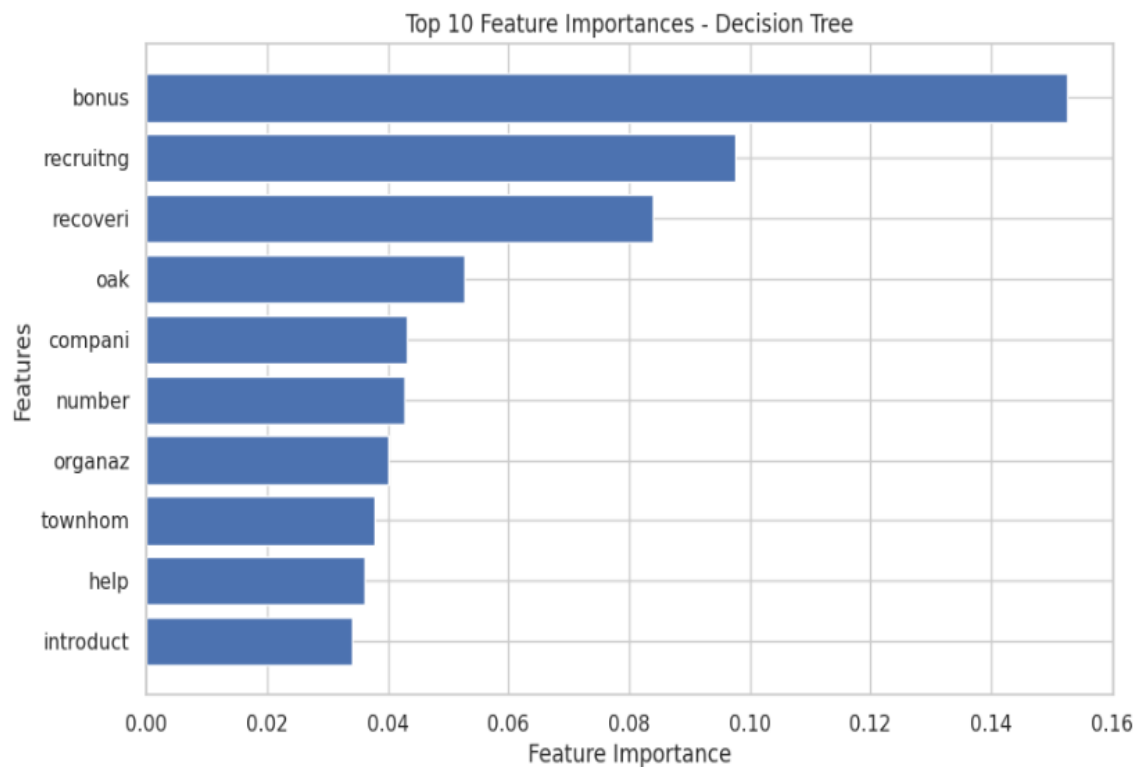
After applying SMOTE, the recall for fraudulent job postings improves dramatically, reaching up to 99% in some cases. However, this comes at the expense of reduced overall accuracy and precision. For example, the Decision Tree with SMOTE achieved near-perfect recall, but its precision plummeted, indicating that many legitimate job postings were incorrectly classified as fraudulent (García et al., 2012).

4.4 Feature Importance Analysis of Gradient Boosting and Decision Tree Models

The feature importance analysis of both Gradient Boosting and Decision Tree models revealed key terms that significantly contribute to identifying fraudulent job postings. In the Gradient Boosting model, the most important features included terms like bonus (importance: 0.1506), company (0.0695), and recruiting (0.0632), highlighting the importance of specific job benefits and company-related words in detecting fraud. After applying SMOTE, the most prominent features shifted slightly, with company (0.1956) and work (0.1599) having the highest importance.



Similarly, for the Decision Tree model, words such as bonus (0.1527), recruiting (0.0976), and recovery (0.0840) emerged as key indicators of fraudulent activity. After applying SMOTE, terms like company (0.2228) and "work" (0.1446) were identified as the most important, suggesting that these terms play a critical role in distinguishing between legitimate and fraudulent job postings.



4.5 Cross Validation

In this study, the dataset was split into **training** (80%) and **testing** (20%) sets to evaluate the performance of the machine learning models. The training set was used to fit the models, while the testing set provided an independent evaluation of the models' ability to generalize to unseen data. This method ensured that the results were not biased by overfitting to the training data, allowing us to assess the models' performance in real-world scenarios.

The models were evaluated based on key performance metrics, including accuracy, precision, recall, and F1-score. This train-test split approach helped us gauge how well the models could classify fraudulent and non-fraudulent job postings, ensuring that the models are effective for fraud detection in job listings.

4.6 Best Performing Result

- **Without SMOTE:** The Gradient Boosting Classifier demonstrated the best balance between precision, recall, and F1-score for detecting fraudulent jobs. With an overall accuracy of 96.92% and an F1-score of 0.54 for the fraudulent class, it outperformed other models in detecting fraudulent postings while maintaining reasonable accuracy for non-fraudulent postings.
- **With SMOTE:** The Decision Tree Classifier, with a recall rate of 99% for fraudulent jobs, showed the best performance in detecting fraudulent job ads. However, this improvement in recall

came at the cost of a significant drop in precision, as many legitimate jobs were misclassified as fraudulent, resulting in a lower F1-score.

4.7 Effectiveness and Impact of SMOTE

For the purpose of identifying fake job posts, SMOTE (Synthetic Minority Over-sampling Technique) shown remarkable efficacy in raising recall rates in all models. Many models reached near-perfect recall after applying SMOTE, which indicates that most fake postings were correctly identified. This is especially crucial for fraud detection since it might be expensive to overlook fraudulent situations. Nevertheless, there was a price to be paid for this boost in recall: a large rise in false positives, or job listings that were mistakenly flagged as fake. This resulted in a decrease in precision, which affected the models' overall accuracy.

Although SMOTE effectively reduces class imbalance and improves recollection, the increase in false positives shows how limited it is. This shows that managing unbalanced datasets in fraud detection may not be best served by SMOTE alone. SMOTE may be combined with additional strategies, such as hyperparameter tweaking or more sophisticated unbalanced learning approaches like SMOTE-ENN, to get a better recall-to-precision trade-off. Together, these strategies may decrease false positives while preserving the high recall required for efficient fraud detection.

4.8 Business Oriented Findings

The performance of the machine learning models has direct implications for businesses, especially online job platforms. Models with higher recall rates, such as Gradient Boosting, can significantly reduce the presence of fraudulent postings, enhancing the platform's security and reinforcing user trust. For businesses, this means fewer scam listings slipping through the cracks, reducing the potential for user complaints or legal challenges related to fraud. However, the trade-off for improved recall is often lower precision, which could result in false positives—legitimate job postings being mistakenly flagged as fraudulent. For a business, this creates a risk of alienating genuine employers and losing revenue from potential clients. Therefore, the findings suggest that while high recall is beneficial for security, it must be balanced with precision to avoid discouraging legitimate business activity. Implementing a robust fraud detection system that can handle this trade-off effectively will be essential for job platforms looking to maintain both user trust and business relationships.

Discussion

The rapid rise in fraudulent job advertisements poses a significant challenge to both job seekers and employers. Detecting these fake job ads has become crucial in maintaining the integrity of online job platforms and ensuring the safety of users. This study implemented six different machine learning (ML) models—Decision Tree, Support Vector Classifier (SVC), Random Forest, Naive Bayes, Gradient Boosting, and Logistic Regression—to predict and detect fraudulent job postings. The application of Synthetic Minority Over-sampling Technique (SMOTE) was also explored to address the class imbalance between genuine and fake job advertisements.

5.1 Model Evaluation

5.1.1 Decision Tree Classifier

The Decision Tree classifier yielded an accuracy of **97.01%** without the application of SMOTE. However, the model struggled with detecting fraudulent job postings, achieving a recall of only **0.39** for the fraudulent class. This implies that the model missed a significant number of fraudulent postings despite its high overall accuracy. The classifier was more effective in identifying legitimate job ads, leading to a high precision of **0.97** for the fraudulent class. The F1-score for fraudulent postings was **0.56**, indicating a moderate balance between precision and recall.

When SMOTE was applied to balance the dataset, the model's recall improved dramatically to **0.99**, indicating near-perfect detection of fraudulent postings. However, this improvement came at a significant cost—precision dropped to **0.24**, and overall accuracy fell to **84.48%**. The dramatic increase in false positives explains this trade-off, as the classifier began labeling a large number of legitimate job ads as fraudulent.

These results align with previous studies, which found that Decision Tree models perform well with balanced data but tend to overfit minority classes when SMOTE is applied. While SMOTE successfully boosts recall, the resulting surge in false positives highlights the need for further fine-tuning of the model's hyperparameters to maintain a balance between precision and recall.

5.1.2 Support Vector Classifier (SVC)

The Support Vector Classifier (SVC) without SMOTE achieved an accuracy of **96.87%** and a perfect precision of **1.00** for fraudulent job postings. However, the recall was relatively low, at only **0.35**, which means that the model missed a large portion of fraudulent job ads. The F1-score for fraudulent postings was **0.52**, showing that the model is highly conservative when it comes to classifying ads as fraudulent, but very accurate when it does.

After applying SMOTE, the model's recall for fraudulent postings soared to **0.99**, as with the Decision Tree classifier. However, this also led to a significant drop in precision (**0.24**) and accuracy (**84.48%**). The increased recall came at the expense of precision, as the model began classifying many legitimate job ads as fraudulent.

The behaviour of SVC in this case is consistent with other studies that have found SVC to be effective in handling high-dimensional data like text but prone to precision-recall trade-offs when SMOTE is applied (Srivastava, 2022). Fine-tuning the decision boundary of SVC or using alternative techniques like cost-sensitive learning could help mitigate these trade-offs.

5.1.3 Random Forest Classifier

The Random Forest model performed similarly to the Decision Tree and SVC models, achieving an accuracy of **97.06%** without SMOTE. The model had perfect precision for the fraudulent class (**1.00**) but a recall of only **0.39**. The F1-score for fraudulent postings was **0.56**, indicating a balance between precision and recall that is like the Decision Tree's performance.

With SMOTE, the recall for fraudulent postings improved to **0.99**, but as with the other models, precision dropped to **0.24** and overall accuracy fell to **84.65%**. The Random Forest classifier's ensemble nature generally improves robustness and reduces overfitting, but it faces the same challenges when

dealing with imbalanced data, as many legitimate ads are classified as fraudulent after SMOTE application.

Previous studies have found Random Forest to be highly effective in handling imbalanced datasets, but the results indicate that using SMOTE without proper tuning can lead to significant declines in performance (Byeon, 2021; Zhou et al., 2022; Ismail, 2023). A more nuanced approach, such as adjusting the class weights or employing an adaptive version of SMOTE, could yield better results.

5.1.4 Naive Bayes Classifier

Naive Bayes performed the worst among all models, with an accuracy of **95.44%** and a recall of only **0.10** for the fraudulent class. While the precision for fraudulent postings was **0.71**, the model's low recall (i.e., detecting only 10% of fraudulent postings) makes it a poor choice for this task. The F1-score was **0.17**, indicating a significant imbalance between precision and recall.

Naive Bayes struggles with complex patterns in text data because it assumes independence between features, which is rarely the case in real-world data (Shang, 2018; Hulu & Sihotang, 2022; Yadav & Thareja, 2019). Additionally, SMOTE was not applied to Naive Bayes because the algorithm relies on probability estimates, which can be skewed by synthetic data points generated by SMOTE (Wilk et al., 2016). As such, Naive Bayes is not an ideal model for detecting fraudulent job ads in this context.

5.1.5 Gradient Boosting Classifier

The Gradient Boosting classifier without SMOTE achieved an accuracy of **96.92%**, a precision of **0.98**, and a recall of **0.37** for fraudulent postings. The F1-score was **0.54**, indicating a fairly balanced performance in detecting fraudulent job ads. Gradient Boosting, known for its ability to handle imbalanced data by learning from previous errors, performed similarly to the Random Forest model, albeit with slightly better recall.

When SMOTE was applied, the recall improved to **0.99**, but precision dropped to **0.22**, and accuracy fell to **82.63%**. The model, like others, began misclassifying legitimate job ads as fraudulent, leading to a significant increase in false positives.

Gradient Boosting models are often favoured for their ability to mitigate overfitting and capture complex relationships in the data (Natekin & Knoll, 2013). However, as seen here, SMOTE application can lead to precision-recall trade-offs. Tuning hyperparameters or exploring different boosting techniques (e.g., XGBoost) could potentially improve performance without sacrificing precision.

5.1.6 Logistic Regression

Logistic Regression performed similarly to the other models without SMOTE, achieving an accuracy of **96.39%**, with a precision of **1.00** and a recall of **0.23** for fraudulent postings. The F1-score for fraudulent postings was **0.38**, indicating that while the model is highly accurate in identifying fraudulent ads when it does, it fails to detect a substantial portion of them.

Like Naive Bayes, Logistic Regression struggles to balance precision and recall for the minority class without specialized techniques such as SMOTE. While it remains effective for binary classification tasks, its inability to detect many fraudulent postings makes it less suitable for this context without further optimization. Previous studies have also highlighted the comparative performance of SVM against

logistic regression, showcasing that SVM can often outperform logistic regression in terms of classification accuracy when applied to job ad datasets (Vidros et al., 2017).

5.2 Insights from Feature Importance in Fraud Detection

The feature importance results from both the Gradient Boosting and Decision Tree models provide valuable insights into how specific terms influence fraud detection in job postings. Notably, terms like bonus, company, and work consistently appear as high-impact features across multiple models. This suggests that fraudulent job ads may often focus on excessive benefits (e.g., bonus) and vague company descriptions (e.g., company), raising red flags for fraud detection systems. Furthermore, as Mahbub et al. (2022) says, the importance of words such as recruiting, and recovery indicates the significance of specific terminology or even misspellings in identifying potentially fraudulent ads.

The application of SMOTE notably increased the prominence of words like company and work in both models, reinforcing the idea that fraudulent job postings may consistently focus on these general terms. This feature analysis provides actionable insights for job platforms to refine their detection algorithms by focusing on these high-impact terms and patterns.

5.3 Improving Model Performance

The findings from this study suggest several avenues for improving model performance:

1. **Hyperparameter Tuning:** Fine-tuning hyperparameters, particularly for ensemble models like Random Forest and Gradient Boosting, can help strike a better balance between precision and recall (Laila et al., 2022). Techniques such as **grid search** or **random search** can be employed to identify the optimal settings for model performance.
2. **Alternative Sampling Techniques:** While SMOTE improves recall, it also introduces a high number of false positives. Alternatives like **ADASYN (Adaptive Synthetic Sampling)** or **SMOTE-ENN** can be explored to reduce false positives while maintaining high recall. **Cost-sensitive learning** is another technique that can be used to directly address the imbalance between legitimate and fraudulent postings (Chawla et al., 2002; Saputra & Wijayanto, 2021).
3. **Class Weight Adjustment:** Adjusting class weights in algorithms such as Random Forest and SVC can help address data imbalance without relying on oversampling techniques like SMOTE. This method can improve recall for the minority class (fraudulent jobs) while maintaining precision.
4. **Threshold Tuning:** Adjusting the decision threshold for classifiers like SVM, Logistic Regression, and Gradient Boosting can help optimize the trade-off between precision and recall. Depending on whether false positives or false negatives are a greater concern, the threshold can be modified to meet the specific goals of the model.
5. **Feature Engineering:** Further feature engineering, including techniques like **n-grams** and domain-specific feature extraction (e.g., fraud-related keywords), can enhance the model's ability to capture fraudulent patterns in job descriptions (Ikeda et al., 2020). Incorporating external data such as company profiles or job market trends can also improve classification accuracy by making the model more context aware.

Conclusion

This study aimed to detect fraudulent job postings using various machine learning models, focusing on improving recall for the minority class while balancing overall accuracy and precision. Six machine learning algorithms—Decision Tree, Support Vector Classifier (SVC), Random Forest, Naive Bayes, Gradient Boosting, and Logistic Regression—were evaluated for their performance in predicting fraudulent job postings. The dataset was highly imbalanced, with real job postings significantly outnumbering fraudulent ones. To address this imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied, which dramatically improved recall but introduced a trade-off in terms of precision, leading to more false positives.

The study found that without SMOTE, most models achieved high overall accuracy but struggled with recall for fraudulent postings, meaning they failed to detect a substantial number of fraudulent job ads. The Gradient Boosting classifier, without SMOTE, provided the best balance between precision, recall, and F1-score, making it the most effective model overall. However, after applying SMOTE, the Decision Tree classifier performed best in terms of recall, achieving near-perfect detection of fraudulent job postings. Despite this, the improvement in recall came at the cost of precision, resulting in a significant increase in false positives, which decreased the overall reliability of the model for real-world applications.

The findings demonstrate that while SMOTE is effective in increasing recall, it also causes a notable rise in false positives, which could be problematic for job platforms that rely on accurate classification. Therefore, SMOTE alone may not be sufficient to create an ideal fraud detection model. Instead, a combination of techniques such as hyperparameter tuning, advanced sampling methods like SMOTE-ENN, and class weight adjustments can provide a more balanced approach to improving both recall and precision.

Further, natural language processing (NLP) played a key role in extracting valuable insights from the job descriptions, company profiles, and other text-based features. Feature engineering, such as extracting fraud-related keywords or using n-grams, could enhance the detection of fraudulent patterns, making the models more robust.

In conclusion, while machine learning models provide powerful tools for detecting fraudulent job postings, achieving a balance between recall and precision remains a challenge, especially with imbalanced datasets. Future work should focus on combining SMOTE with more advanced techniques to minimize false positives while maintaining high recall. Improving interpretability through models like SHAP could also help platform administrators and policymakers better understand the factors contributing to fraud detection, leading to safer and more secure job platforms.

Recommendations

Based on the findings of this study, several strategies are recommended to improve the detection of fraudulent job postings. First, hyperparameter tuning should be prioritized, especially for ensemble models like Random Forest and Gradient Boosting, to strike a better balance between precision and recall. Grid search or random search can help identify optimal settings for model performance. Second, alternative sampling techniques such as ADASYN or SMOTE-ENN should be explored to reduce false positives while maintaining high recall, especially for imbalanced datasets. Third, class weight adjustments in models like SVC and Random Forest can address imbalance without relying solely on oversampling methods. Fourth, further feature engineering, including domain-specific keywords and n-grams, should be implemented to enhance the model's ability to identify fraudulent patterns in job descriptions. Finally, using threshold tuning for classifiers can help optimize the trade-off between precision and recall based on specific needs, such as minimizing false positives.

Limitations in this study include the use of traditional machine learning models, which may not capture deeper complexities compared to advanced techniques like deep learning. Feature importance analysis lacked deeper interpretability tools such as SHAP. Lastly, the study's scope was limited to text-based features, omitting potential valuable metadata like user behaviour.

To implement machine learning models effectively in their fraud detection systems, online job platforms should consider integrating these models into their existing screening processes. A hybrid approach that combines machine learning with human moderation can be particularly effective, allowing businesses to capture the nuanced patterns that machines might miss. It is also important for businesses to continuously update and retrain these models as fraudulent tactics evolve. Furthermore, platforms should prioritize precision in cases where false positives are a significant risk—such as when dealing with high-volume legitimate business clients—while emphasizing recall in scenarios where platform integrity is at stake, such as during high-profile recruitment drives. By carefully balancing security needs with operational efficiency, platforms can mitigate the risks associated with fraudulent job postings while maintaining a seamless experience for legitimate businesses. Additionally, companies should evaluate the economic costs of false positives versus the risks of fraud slipping through, tailoring their fraud detection systems to align with their strategic priorities.

References

- Assiri, A., Nazir, S., & Velastín, S. (2020). Breast tumor classification using an ensemble machine learning method. *Journal of Imaging*, 6(6), 39. <https://doi.org/10.3390/jimaging6060039>
- Bakasa, W. (2023). Stacked ensemble deep learning for pancreas cancer classification using extreme gradient boosting. *Frontiers in Artificial Intelligence*, 6. <https://doi.org/10.3389/frai.2023.1232640>
- Baki, S., Verma, R., & Gnawali, O. (2020). Scam augmentation and customization: identifying vulnerable users and arming defenders.. <https://doi.org/10.1145/3320269.3384753>
- BBC News (2021) UK job seeker defrauded by fake job posting. Available at: <https://www.bbc.co.uk> (Accessed: 11 September 2024).
- Bekkerman, R. and Gavish, M. (2011). High-precision phrase-based document classification on a modern scale.. <https://doi.org/10.1145/2020408.2020449>
- Biau, G. and Scornet, E. (2016). A random forest guided tour. *Test*, 25(2), 197-227. <https://doi.org/10.1007/s11749-016-0481-7>
- Blake, R., Kyper, E., & Douglas, M. (2012). Using business intelligence for operational decision-making in call centers. *International Journal of Decision Support System Technology*, 4(1), 43-54. <https://doi.org/10.4018/jdsst.2012010104>
- Byeon, H. (2021). Exploring parkinson's disease predictors based on basic intelligence quotient and executive intelligence quotient. *International Journal of Advanced Computer Science and Applications*, 12(4). <https://doi.org/10.14569/ijacsa.2021.0120414>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). Smote: synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321-357. <https://doi.org/10.1613/jair.953>
- Chen, H. and Fu, D. (2018). An improved naive bayes classifier for large scale text.. <https://doi.org/10.2991/icaite-18.2018.9>
- Dastjerdi, A., Foroghi, D., & Kiani, G. (2019). Detecting manager's fraud risk using text analysis: evidence from iran. *Journal of Applied Accounting Research*, 20(2), 154-171. <https://doi.org/10.1108/jaar-01-2018-0016>
- Erol, G., Uzbaş, B., Yücelbaş, C., & Yücelbaş, Ş. (2022). Analyzing the effect of data preprocessing techniques using machine learning algorithms on the diagnosis of covid-19. *Concurrency and Computation Practice and Experience*, 34(28). <https://doi.org/10.1002/cpe.7393>
- Faraji, Z. (2022). A review of machine learning applications for credit card fraud detection with a case study. *Seisense Journal of Management*, 5(1), 49-59. <https://doi.org/10.33215/sjom.v5i1.770>
- Feng, D., Wang, H., Li, L., Guo, Y., Bissyandé, T., Liu, T., ... & Klein, J. (2018). Frauddroid: automated ad fraud detection for android apps.. <https://doi.org/10.1145/3236024.3236045>
- Fernando, J., Budiraharjo, R., & Haganusa, E. (2019). Spam classification on 2019 indonesian president election youtube comments using multinomial naïve-bayes. *Indonesian Journal of Artificial Intelligence and Data Mining*, 2(1). <https://doi.org/10.24014/ijaidm.v2i1.6445>
- García, V., Sánchez, J., Martín-Félez, R., & Mollineda, R. (2012). Surrounding neighborhood-based smote for learning from imbalanced data sets. *Progress in Artificial Intelligence*, 1(4), 347-362. <https://doi.org/10.1007/s13748-012-0027-5>
- Grant-Smith, D. (2022). Key trends in employment scams in Australia: what are the gaps in knowledge about recruitment fraud?.. <https://doi.org/10.5204/book.eprints.228500>
- Gurusami, S. (2018). The carceral web we weave: carceral citizens' experiences of digital punishment and solidarity. *Punishment & Society*, 21(4), 435-453. <https://doi.org/10.1177/1462474518790237>
- Hanif, A. (2024). Machine learning approach in predicting fraudulent job advertisement. *International Journal of Academic Research in Business and Social Sciences*, 14(1). <https://doi.org/10.6007/ijarbss/v14-i1/20532>

- Hansraj, A., Adeliyi, T., & Wing, J. (2021). Detection of online fake news using blending ensemble learning. *Scientific Programming*, 2021, 1-10. <https://doi.org/10.1155/2021/3434458>
- Han, H. and Jiang, X. (2014). Overcome support vector machine diagnosis overfitting. *Cancer Informatics*, 13s1, CIN.S13875. <https://doi.org/10.4137/cin.s13875>
- Hulu, C. and Sihotang, H. (2022). Plant disease diagnosis expert system cardamom (amomum cardamomum L.) using the naive bayes method web-based. *Jurnal Mandiri It*, 10(2), 68-73. <https://doi.org/10.35335/mandiri.v10i2.94>
- Ikeda, C., Ouazzane, K., & Yu, Q. (2020). A new framework of feature engineering for machine learning in financial fraud detection. *Computer Science & Information Technology (CS & IT)*. <https://doi.org/10.5121/csit.2020.101517>
- International Labour Organization (2022) World Employment and Social Outlook: Trends 2022. Available at: <https://www.ilo.org> (Accessed: 11 September 2024)
- Ismail, E. (2023). A hybrid stacking-smote model for optimizing the prediction of autistic genes. *BMC Bioinformatics*, 24(1). <https://doi.org/10.1186/s12859-023-05501-y>
- Jiang, X. (2022). Toshiba and luckin coffee: a study of the reasons for committing financial fraud. *Proceedings of Business and Economic Studies*, 5(5), 169-173. <https://doi.org/10.26689/pbes.v5i5.4423>
- Jiang, X. (2024). Application of decision tree and machine learning in new energy vehicle maintenance decision making. *Applied Mathematics and Nonlinear Sciences*, 9(1). <https://doi.org/10.2478/amns-2024-1463>
- Kadam, D. (2024). Machine learning approaches to credit card fraud detection. *International Journal for Research in Applied Science and Engineering Technology*, 12(4), 2802-2807. <https://doi.org/10.22214/ijraset.2024.60531>
- Kassem, R. (2023). Mapping romance fraud research – a systematic review. *Journal of Financial Crime*, 31(4), 974-992. <https://doi.org/10.1108/jfc-06-2023-0160>
- Kharraz, A., Robertson, W., & Kirda, E. (2018). Surveylance: automatically detecting online survey scams.. <https://doi.org/10.1109/sp.2018.00044>
- Kh.TQ, M. (2024). Classification of tuberculosis based on chest x-ray images for imbalance data using smote. *International Journal of Computing and Digital Systems*, 15(1), 981-993. <https://doi.org/10.12785/ijcds/160171>
- Kucuk, E. (2024). Comparative analysis of machine learning algorithms for biomedical text document classification: a case study on cancer-related publications. *Medicine Science | International Medical Journal*, 13(1), 171. <https://doi.org/10.5455/medscience.2023.10.209>
- Kudo, W., Nishiguchi, M., & Toriumi, F. (2019). Fraudulent user detection on rating networks based on expanded balance theory and gcns.. <https://doi.org/10.1145/3341161.3342929>
- Laila, U., Mahboob, K., Khan, A. W., Khan, F., & Taekeun, W. (2022). An ensemble approach to predict early-stage diabetes risk using machine learning: an empirical study. *Sensors*, 22(14), 5247. <https://doi.org/10.3390/s22145247>
- Lin, D. (2023). An empirical analysis of machine learning for fraud detection in diverse financial scenarios. *Advances in Economics Management and Political Sciences*, 42(1), 202-216. <https://doi.org/10.54254/2754-1169/42/20232110>
- Lukersmith, S., Taylor, J., & Salvador-Carulla, L. (2021). Vagueness and ambiguity in communication of case management: a content analysis in the Australian national disability insurance scheme. *International Journal of Integrated Care*, 21(1), 17. <https://doi.org/10.5334/ijic.5590>
- McCann, J., Stockton, D., & Godward, S. (2002). Impact of false-positive mammography on subsequent screening attendance and risk of cancer. *Breast Cancer Research*, 4(5). <https://doi.org/10.1186/bcr455>
- Mahbub, S., Pardede, E., & Kayes, A. S. M. (2022). Online recruitment fraud detection: a study on contextual features in Australian job industries. *IEEE Access*, 10, 82776-82787. <https://doi.org/10.1109/access.2022.3197225>

- Mehta, K., Rajagopal, N., Gaikwad, S., & Yadav, S. (2022). Machine learning techniques for true and fake job posting. *Ecs Transactions*, 107(1), 2383-2389. <https://doi.org/10.1149/10701.2383ecst>
- Mijwil, M. and Abttan, R. (2021). Utilizing the genetic algorithm to pruning the c4.5 decision tree algorithm. *Asian Journal of Applied Sciences*, 9(1). <https://doi.org/10.24203/ajas.v9i1.6503>
- MoDastoni, D. (2023). Exploring methods to make ai decisions more transparent and understandable for humans. *AEI*, 3(1). <https://doi.org/10.54254/2977-3903/3/2023037>
- Mutemi, A. and Bação, F. (2023). A numeric-based machine learning design for detecting organized retail fraud in digital marketplaces. *Scientific Reports*, 13(1). <https://doi.org/10.1038/s41598-023-38304-5>
- Natekin, A. and Knoll, A. (2013) Gradient Boosting Machines, a tutorial, *Frontiers*. Available at: <https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2013.00021/full> (Accessed: 08 September 2024).
- Office for National Statistics (2023) Labour market overview, UK: July 2023. Available at: <https://www.ons.gov.uk> (Accessed: 11 September 2024).
- Praveen, A., Harsita, R., Murali, R., & Niveditha, S. (2023). Detecting fake job posting using ml classifications and ensemble model. *Advances in Science and Technology*. <https://doi.org/10.4028/p-hdm12o>
- Rahman, H., Wah, Y., & Ong, S. (2021). Predictive performance of logistic regression for imbalanced data with categorical covariate. *Pertanika Journal of Science and Technology*, 29(1). <https://doi.org/10.47836/pjst.29.1.10>
- Rath, Y., Malcolm, J., Michailovich, O., Goldstein, J., Seidman, L., McCarley, R., ... & Shenton, M. (2010). Biomarkers for identifying first-episode schizophrenia patients using diffusion weighted imaging., 657-665. https://doi.org/10.1007/978-3-642-15705-9_80
- Ravenelle, A., Janko, E., & Kowalski, K. (2022). Good jobs, scam jobs: detecting, normalizing, and internalizing online job scams during the covid-19 pandemic. *New Media & Society*, 24(7), 1591-1610. <https://doi.org/10.1177/14614448221099223>
- Raza, A., Ubaid, S., Younas, F., & Akhtar, F. (2022). Fake e job posting prediction based on advance machine learning approachs. *International Journal of Research Publication and Reviews*, 689-695. <https://doi.org/10.55248/gengpi.2022.3.2.7>
- Rofik, R. (2024). Optimization of svm and gradient boosting models using gridsearchcv in detecting fake job postings. *Matrik Jurnal Manajemen Teknik Informatika Dan Rekayasa Komputer*, 23(2), 419-430. <https://doi.org/10.30812/matrik.v23i2.3566>
- Sandhu, N. (2019). Behavioural red flags of fraud: an ex post assessment of types and frequencies. *Global Business Review*, 21(2), 507-525. <https://doi.org/10.1177/0972150919850410>
- Saputra, A. M. W. and Wijayanto, A. W. (2021). Implementation of ensemble techniques for diarrhea cases classification of under-five children in indonesia. *JITK (Jurnal Ilmu Pengetahuan Dan Teknologi Komputer)*, 6(2), 175-180. <https://doi.org/10.33480/jitk.v6i2.1935>
- Shang, S. (2018). An improved text sentiment analysis algorithm based on tf-gini. *IJPE*. <https://doi.org/10.23940/ijpe.18.09.p8.20082014>
- Shim, J. and Seok, K. (2012). Semiparametric kernel logistic regression with longitudinal data. *Journal of the Korean Data and Information Science Society*, 23(2), 385-392. <https://doi.org/10.7465/jkdi.2012.23.2.385>
- Sipayung, E., Yanti, H., & Setya, A. (2023). Impact of anti-fraud awareness, fraud detection procedures, and technology to fraud detection skill., 783-787. https://doi.org/10.2991/978-2-494069-49-7_132
- Sitepu, R. (2022). The analysis of support vector machine (svm) on monthly covid-19 case classification. *International Journal on Information and Communication Technology (Ijoict)*, 8(2), 40-52. <https://doi.org/10.21108/ijoict.v8i2.671>
- Smity, S., Soui, M., & Ghedira, K. (2022). Tri-xgboost model: an interpretable semi-supervised approach for addressing bankruptcy prediction.. <https://doi.org/10.22541/au.166671556.62164357/v1>

- Srivastava, R. (2022). Identification of online recruitment fraud (orf) through predictive models. EJBESS, 1(1), 42-54. <https://doi.org/10.54878/epv1hr57>
- Turner, C., Jacobs, A., Marques, C., Oates, J., Kamen, D., Anderson, P, ... & Obeid, J. (2017). Word2vec inversion and traditional text classifiers for phenotyping lupus. BMC Medical Informatics and Decision Making, 17(1). <https://doi.org/10.1186/s12911-017-0518-1>
- Ullah, Z. and Jamjoom, M. (2023). A smart secured framework for detecting and averting online recruitment fraud using ensemble machine learning techniques. Peerj Computer Science, 9, e1234. <https://doi.org/10.7717/peerj-cs.1234>
- Vidros, S., Kolias, C., Kambourakis, G., & Akoglu, L. (2017). Automatic detection of online recruitment frauds: characteristics, methods, and a public dataset. Future Internet, 9(1), 6. <https://doi.org/10.3390/fi9010006>
- Wibawa, A., Kurniawan, A., Murti, D., Adiperkasa, R., Putra, S., Kurniawan, S., ... & Nugraha, Y. (2019). Naïve bayes classifier for journal quartile classification. International Journal of Recent Contributions From Engineering Science & It (Ijes), 7(2), 91. <https://doi.org/10.3991/ijes.v7i2.10659>
- Wilk, S., Stefanowski, J., Wojciechowski, S., Farion, K., & Michalowski, W. (2016). Application of preprocessing methods to imbalanced clinical data: an experimental study., 503-515. https://doi.org/10.1007/978-3-319-39796-2_41
- Xia, H., An, W., & Zhang, Z. (2023). Credit risk models for financial fraud detection. Journal of Database Management, 34(1), 1-20. <https://doi.org/10.4018/jdm.321739>
- Yadav, K. and Thareja, R. (2019). Comparing the performance of naive bayes and decision tree classification using r. International Journal of Intelligent Systems and Applications, 11(12), 11-19. <https://doi.org/10.5815/ijisa.2019.12.02>
- Zhang, Z. (2016). Naïve bayes classification in r. Annals of Translational Medicine, 4(12), 241-241. <https://doi.org/10.21037/atm.2016.03.38>
- Zhou, R., Yin, W., Li, W., Ying-chun, W., Lu, J., Li, Z., ... & Hu, X. (2022). Prediction model for infectious disease health literacy based on synthetic minority oversampling technique algorithm. Computational and Mathematical Methods in Medicine, 2022, 1-6. <https://doi.org/10.1155/2022/8498159>

Appendix A: Python Results

- Data Exploration

	job_id	company_profile_length	description_length
count	17880.000000	17880.000000	17880.000000
mean	8940.500000	92.553579	170.445973
std	5161.655742	85.487278	123.304236
min	1.000000	0.000000	0.000000
25%	4470.750000	21.000000	87.000000
50%	8940.500000	83.000000	146.000000
75%	13410.250000	132.000000	224.000000
max	17880.000000	885.000000	2115.000000

	requirements_length	benefits_length
count	17880.000000	17880.000000
mean	78.032942	29.990884
std	81.652956	49.755790
min	0.000000	0.000000
25%	19.000000	0.000000
50%	62.000000	6.000000
75%	108.000000	41.000000
max	1645.000000	694.000000

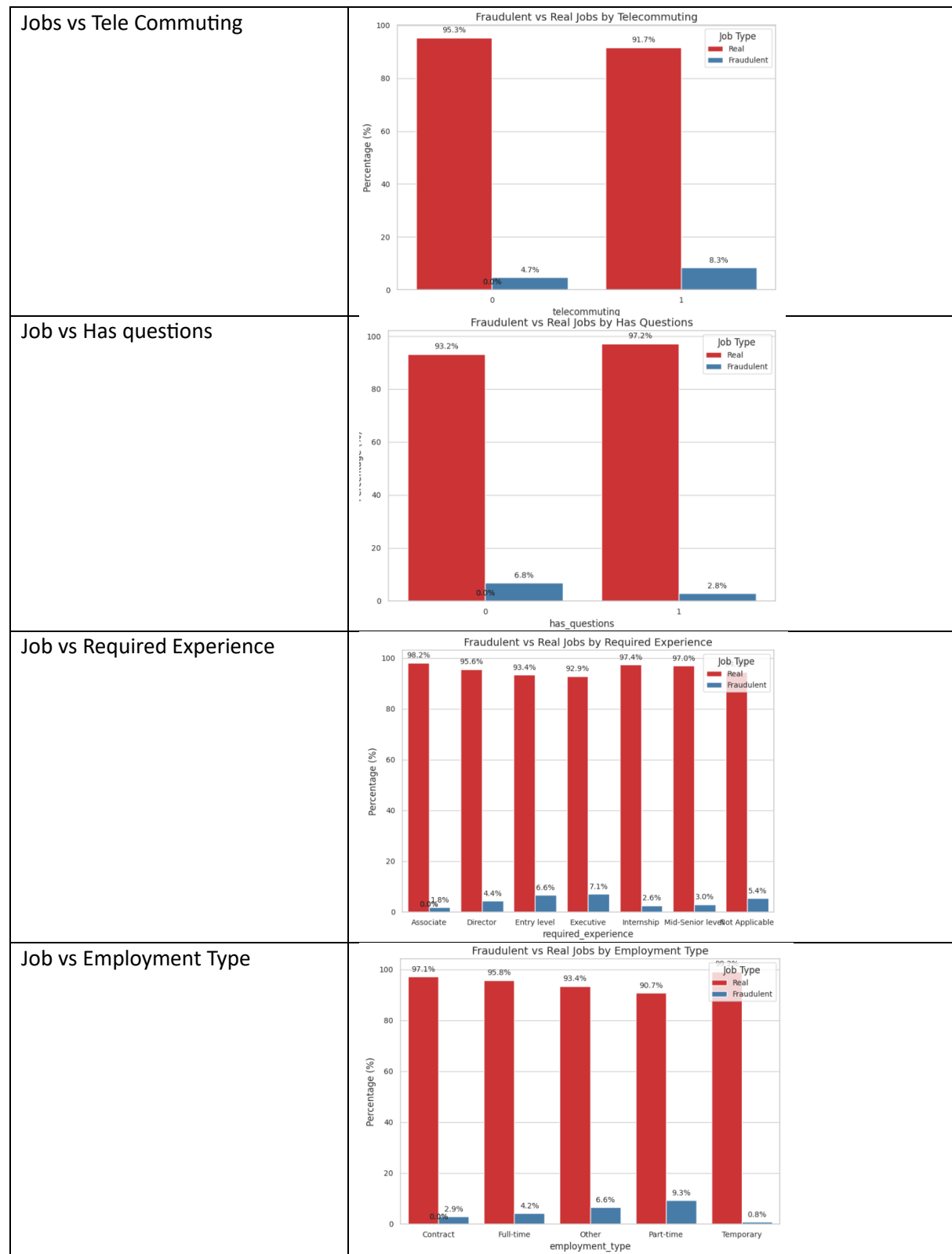
- Unique Values and Data Types

#	Column	Non-Null Count	Dtype
0	job_id	17880 non-null	int64
1	title	17880 non-null	object
2	location	17534 non-null	object
3	department	6333 non-null	object
4	salary_range	2868 non-null	object
5	company_profile	14572 non-null	object
6	description	17879 non-null	object
7	requirements	15184 non-null	object
8	benefits	10668 non-null	object
9	telecommuting	17880 non-null	category
10	has_company_logo	17880 non-null	category
11	has_questions	17880 non-null	category
12	employment_type	14409 non-null	category
13	required_experience	10830 non-null	category
14	required_education	9775 non-null	category
15	industry	12977 non-null	object
16	function	11425 non-null	object
17	fraudulent	17880 non-null	category
18	Cleaned_company_profile	17880 non-null	object
19	Cleaned_description	17880 non-null	object
20	Cleaned_requirements	17880 non-null	object
21	Cleaned_benefits	17880 non-null	object
22	company_profile_length	17880 non-null	int64
23	description_length	17880 non-null	int64
24	requirements_length	17880 non-null	int64
25	benefits_length	17880 non-null	int64

- Missing Values

None	
location	346
department	11547
salary_range	15012
company_profile	3308
description	1
requirements	2696
benefits	7212
industry	4903
function	6455

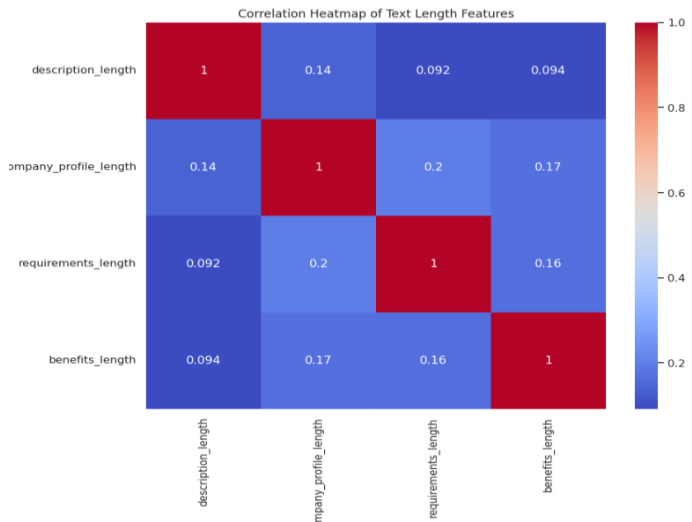
Appendix B: Python Charts



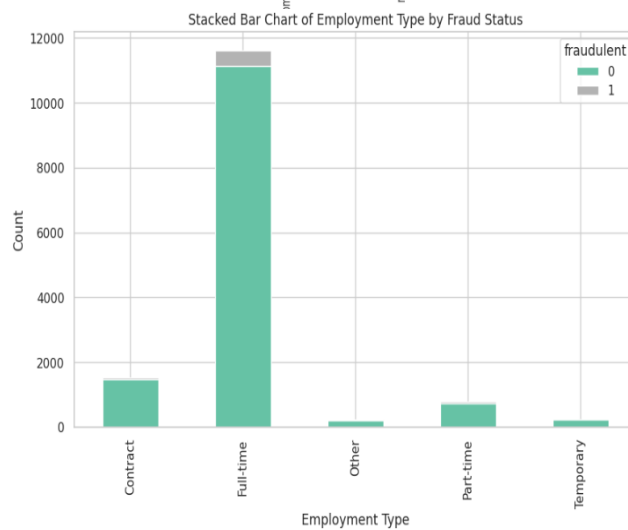
Job vs Description Length



Correlation Heat map



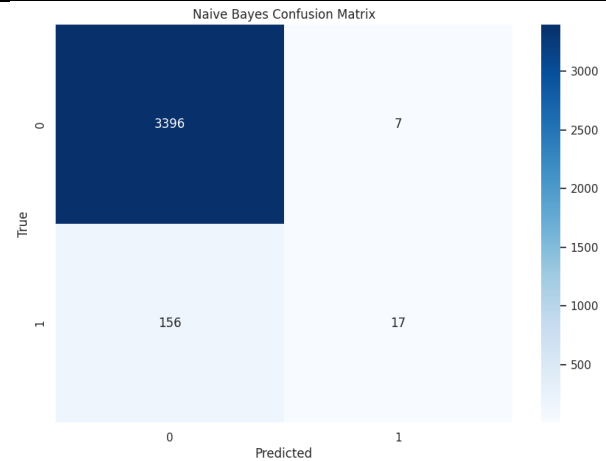
Job vs Employment Status



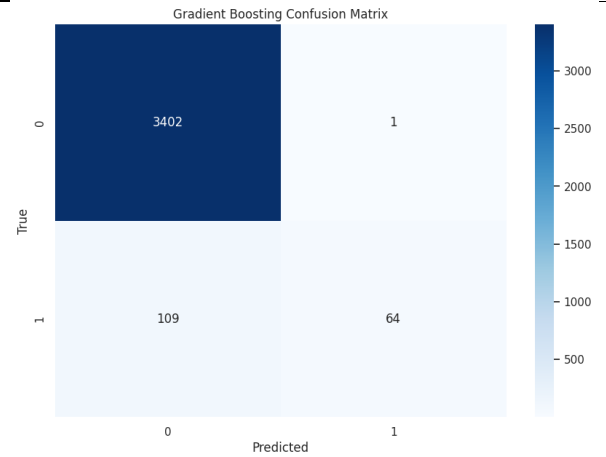
Average Word count for fraud and real jobs for description, requirements and company profile.



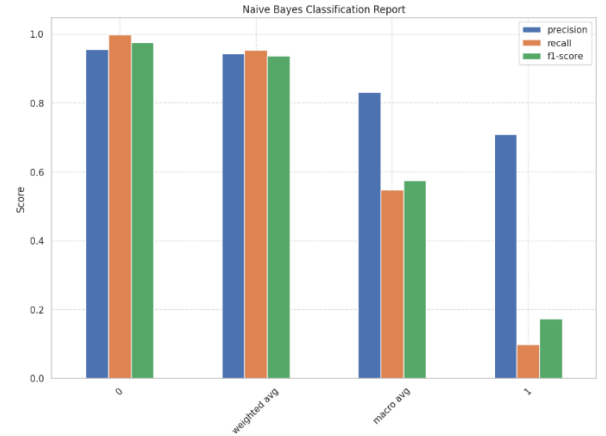
Naïve Bayes Confusion Matrix



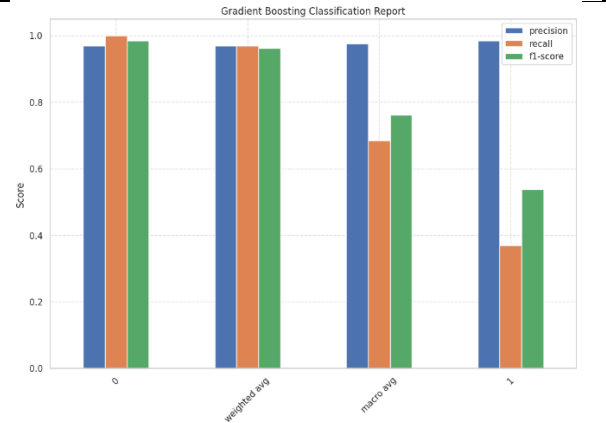
Gradient Boost Confusion Matrix



Naïve Bayes Classification Report



Gradient Boost Classification Report



Appendix C: ML Model Results

- Linear Regression

Logistic Regression Confusion Matrix:

```
[[3395   0]
 [ 139  42]]
```

Logistic Regression Classification Report:

	precision	recall	f1-score	support
0	0.96	1.00	0.98	3395
1	1.00	0.23	0.38	181
accuracy			0.96	3576
macro avg	0.98	0.62	0.68	3576
weighted avg	0.96	0.96	0.95	3576

- Decision Tree (With/Without Smote)

Decision Tree Classifier Results:

Confusion Matrix:

```
[[3401   2]
 [ 105  68]]
```

Accuracy: 0.9701

Classification Report:

	precision	recall	f1-score	support
0	0.97	1.00	0.98	3403
1	0.97	0.39	0.56	173
accuracy			0.97	3576
macro avg	0.97	0.70	0.77	3576
weighted avg	0.97	0.97	0.96	3576

Decision Tree Classifier with SMOTE Results:

Confusion Matrix:

```
[[2850  553]
 [    2  171]]
```

Accuracy: 0.8448

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.84	0.91	3403
1	0.24	0.99	0.38	173
accuracy			0.84	3576
macro avg	0.62	0.91	0.65	3576
weighted avg	0.96	0.84	0.89	3576

- **SVC (with/without Smote)**

Support Vector Classifier Results:

Confusion Matrix:

```
[[3403  0]
 [ 112  61]]
```

Accuracy: 0.9687

Classification Report:

	precision	recall	f1-score	support
0	0.97	1.00	0.98	3403
1	1.00	0.35	0.52	173
accuracy			0.97	3576
macro avg	0.98	0.68	0.75	3576
weighted avg	0.97	0.97	0.96	3576

Support Vector Classifier with SMOTE Results:

Confusion Matrix:

```
[[2850  553]
 [   2  171]]
```

Accuracy: 0.8448

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.84	0.91	3403
1	0.24	0.99	0.38	173
accuracy			0.84	3576
macro avg	0.62	0.91	0.65	3576
weighted avg	0.96	0.84	0.89	3576

- **Random Forrest Classifier (With/without Smote)**

Random Forest Classifier Results:

Confusion Matrix:

```
[[3403  0]
 [ 105  68]]
```

Accuracy: 0.9706

Classification Report:

	precision	recall	f1-score	support
0	0.97	1.00	0.98	3403
1	1.00	0.39	0.56	173
accuracy			0.97	3576
macro avg	0.99	0.70	0.77	3576
weighted avg	0.97	0.97	0.96	3576

Random Forest Classifier with SMOTE Results:

Confusion Matrix:

```
[[2856  547]
 [   2  171]]
```

Accuracy: 0.8465

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.84	0.91	3403
1	0.24	0.99	0.38	173
accuracy			0.85	3576
macro avg	0.62	0.91	0.65	3576
weighted avg	0.96	0.85	0.89	3576

Appendix D: Appendix 3 Dissertation Checklist Sheet

A digitally signed copy of this form should be included as an appendix to the dissertation.

Name: Uzair Abdullah Khan

Date Submitted: 13/09/2024

Signature (Digital): Uzair Abdullah Khan

I confirm that my dissertation contains the following prescribed elements:

☐ My dissertation portfolio meets the style requirements set out in the MSc Business Analytics Portfolio Dissertation Handbook including a word count on the front page of each element.

☐ I have reviewed the Turnitin similarity report prior to submission.

☐ My dissertation title captures succinctly the focus of my dissertation

☐ My title page is formatted as prescribed in the MSc Business Analytics Portfolio Dissertation Handbook

☐ The abstract provides a clear and succinct overview of my study

☐ Each element contains a Table of Contents, and List of Figures and Tables (where appropriate)

☐ My dissertation contains a statement of acknowledgement (optional)

☐ The Introduction section of the research report, at a minimum, covers each of the following issues:

- Background to/context of the project
- Research question(s), aim(s) and objectives
- Why the project is necessary/important
- A summary of the Methodology
- Outline of the key findings
- Overview of chapter structure of the remainder of dissertation

☐ The Background section of the research report, at a minimum, covers each of the following issues:

- Synthesises the key technical literature relating to the topic
- Synthesises the key theoretical literature relating to the topic

- The methodology section of the research report:
 - Details the procedures adopted in carrying out the project (e.g. the data source/acquisition, data processing, procedures for maximising rigour and robustness, methods of data analysis etc). Contains ethical considerations and decisions. This section should not duplicate the technical report, which focuses more on the detailed technical choices and steps.
- The findings section of the research report reports the results in detail and provides possible explanations for the various findings
- The discussion section of the research report makes appropriate linkages between the findings and the literature reviewed
- The conclusions section of the research report includes:
 - Conclusions about each research question and/or hypothesis
 - General conclusions about the research problem
 - Implications for theory, for policy and/or management practice
 - Limitations of the research
 - Suggestions for practice and future research
- The technical report, log book, and reflective discussion have each been included.
- The reference list is in alphabetical order and follows the Harvard system
- I have signed and dated the Candidate Declaration